

E7 Introduction to Computer Programming for Scientists and Engineers

Discussion week 8

A brief introduction to linear algebra

The pillars of applied mathematics

- **Calculus** ... the study of change over space and time.
- **Linear algebra** ... the study of linear transformations.
- **Statistics** ... the study of uncertainty.
- **Optimization theory** ... the study of decision making.
- **Numerical methods** ... how to solve problems with a computer.

Basic concepts in linear algebra

- vector space
- vector
- matrix
- linear transformation

- A vector space is a collection of objects (called vectors) that

a) can be **added** together to produce another vector,

$$a, b \in V \quad \Rightarrow \quad a + b \in V$$

b) can be **multiplied by a scalar** to produce another vector,

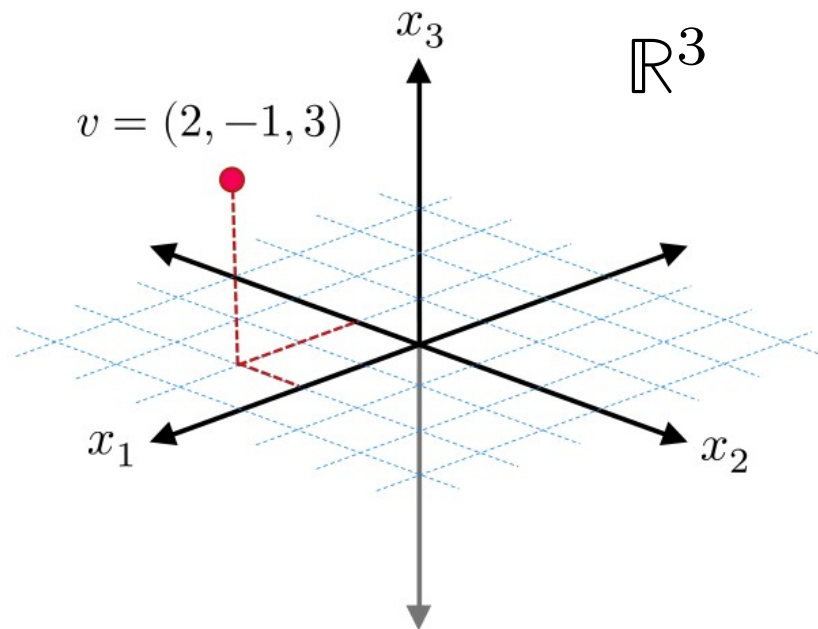
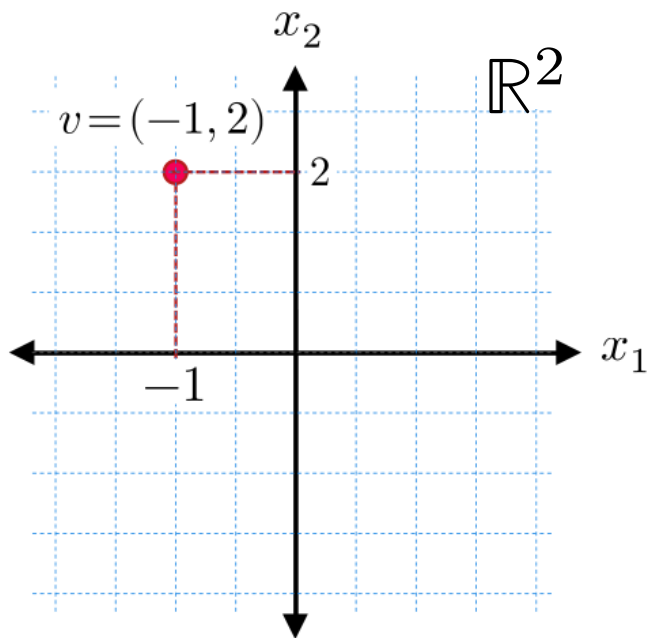
$$\begin{array}{l} a \in V \\ \alpha \in \mathbb{R} \end{array} \quad \Rightarrow \quad \alpha a \in V$$

Examples of vector spaces

- The set of all polynomials of a real variable of degree two or less.
- Numpy arrays of a given shape, with dtype=float.

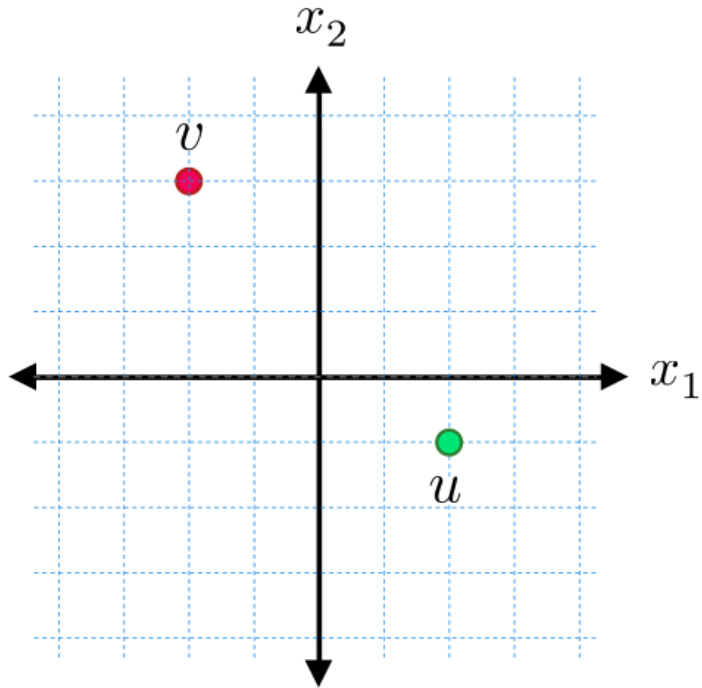
The most important vector spaces: $\mathbb{R}^n$

- For each integer $n \geq 0$, $\mathbb{R}^n$ is a vector space.
- Its vectors are of the form: $v = (v_1, v_2, \dots, v_n)$



Vector addition and scalar multiplication in $\mathbb{R}^2$

With $v = (-2, 3)$ and $u = (2, -1)$, find $u + v$ and $2u$.



Matrices

An m -by- n matrix M (denoted $M \in \mathbb{R}^{m \times n}$) is a grid of numbers with m rows and n columns.

$$M = \begin{bmatrix} m_{11} & m_{12} & m_{13} & m_{14} \\ m_{21} & m_{22} & m_{23} & m_{24} \\ m_{31} & m_{32} & m_{33} & m_{34} \end{bmatrix} \in \mathbb{R}^{3 \times 4}$$

Multiplying a matrix by a vector

- A matrix $M \in \mathbb{R}^{m \times n}$ can be “multiplied” by a vector $v \in \mathbb{R}^p$ when $n = p$.
- That is, the number of columns in M matches the number of items in v .

Example: Can M and v be multiplied?

- $M = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$, $v = (0, 4, 1)$

- $M = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}$, $v = (3, 2, 1)$

How to multiply a matrix by a vector

- With $M \in \mathbb{R}^{m \times n}$ and $v \in \mathbb{R}^n$, their product Mv is a vector in $\mathbb{R}^m$.
- The i 'th element of Mv is the “dot product” of the i 'th row of M with v .
- A *dot product* of two vectors is the sum of their element-wise product.

Example

$$\begin{bmatrix} b_1 \\ b_2 \end{bmatrix} = \begin{bmatrix} m_{11} & m_{12} & m_{13} \\ m_{21} & m_{22} & m_{23} \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix}$$

=

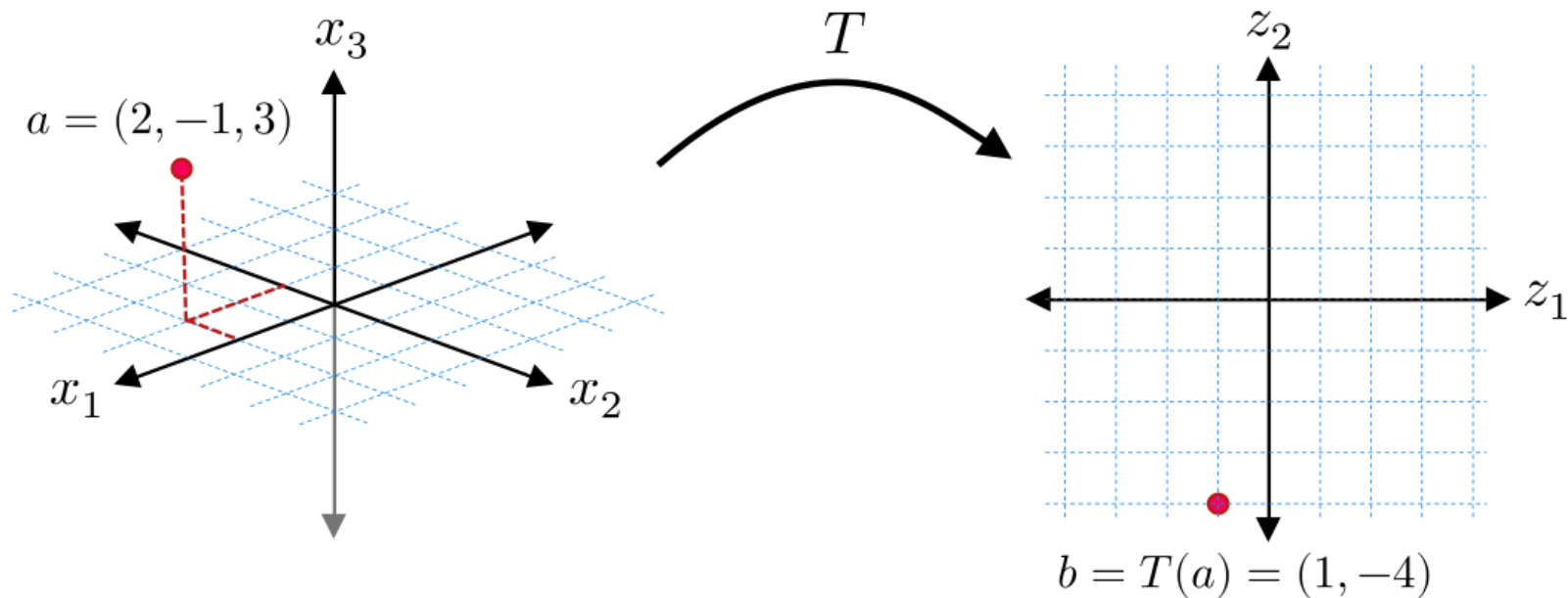
Linear transformations

A *linear transformation* T is function from a vector space V to a vector space W with the property

$$T(u + \alpha v) = T(u) + \alpha T(v) \quad \forall \alpha \in \mathbb{R}, \forall u, v \in V$$

Example $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ maps each $a = (a_1, a_2, a_3)$ to a point $b = (b_1, b_2)$ with

$$T : \begin{cases} b_1 = 3a_1 + 2a_2 - a_3 \\ b_2 = a_2 - a_3 \end{cases}$$

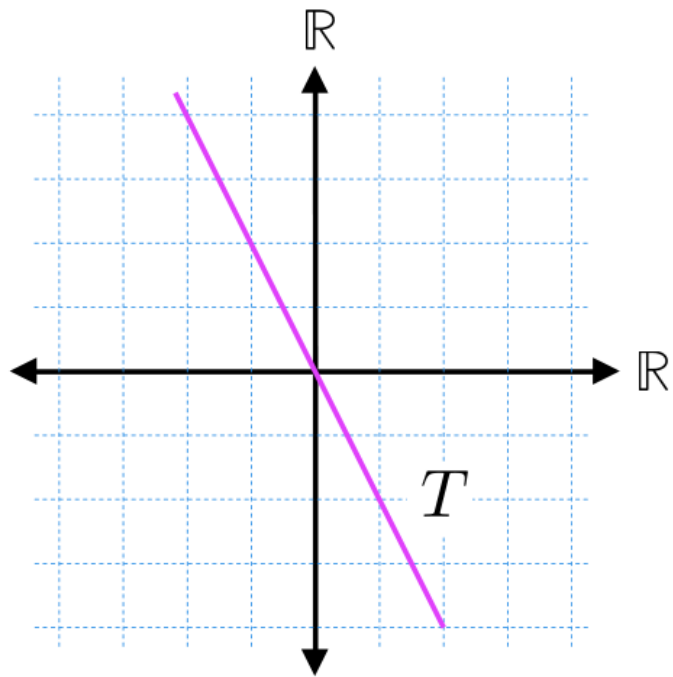


Show that this T is a linear transformation.

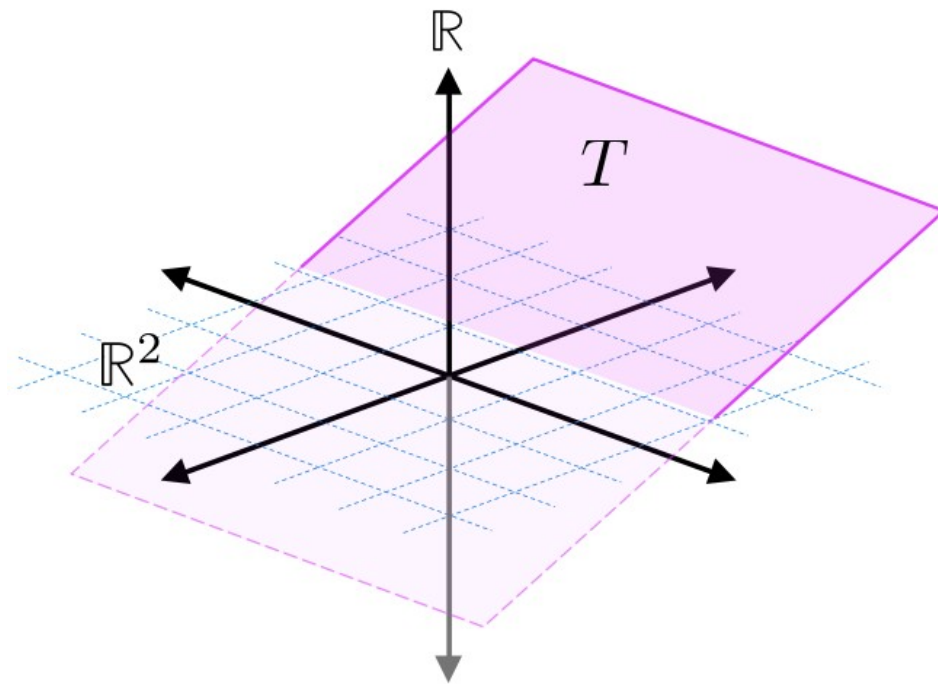
Proof:

Low dimensional linear transformations as a single plot

$$T : \mathbb{R} \rightarrow \mathbb{R}$$



$$T : \mathbb{R}^2 \rightarrow \mathbb{R}$$



Matrix representation theorem

For every linear transformation $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$, there exists a matrix $M \in \mathbb{R}^{m \times n}$, such that

$$T(u) = Mu \quad \forall u \in \mathbb{R}^n$$

(*) Technical caveat: This is true only for “finite dimensional” vector spaces, but this is beyond our scope.

Example

Find the matrix M corresponding to the following linear transformation T .

$$T : \begin{cases} b_1 = 3a_1 + 2a_2 - a_3 \\ b_2 = a_2 - a_3 \end{cases}$$

Applications of linear algebra

- Linear transformations are very common in science and engineering.
 - Linear dynamical systems
 - Statistics and data science
 - Image processing and 3D graphics
 - Electrical circuit analysis
 - etc.
- The study of linear transformations (i.e. linear algebra) includes
 - matrix properties: rank and determinant
 - matrix decompositions: spectral (eigenvalues), singular values (SVD), etc.
 - matrix inversion: Moore-Penrose inverse

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Discussion week 11

Numerical differentiation and integration

Recap

Numerical differentiation

Name	Approximates	Formula	Error bound
Forward difference	$f'(x)$	$\frac{f(x+h) - f(x)}{h}$	$\frac{M_2}{2}h$
Backward difference	$f'(x)$	$\frac{f(x) - f(x-h)}{h}$	$\frac{M_2}{2}h$
Central difference	$f'(x)$	$\frac{f(x+h) - f(x-h)}{2h}$	$\frac{M_4}{6}h^2$
Central difference	$f''(x)$	$\frac{f(x+h) - 2f(x) + f(x-h)}{h^2}$	$\frac{M_4}{12}h^2$

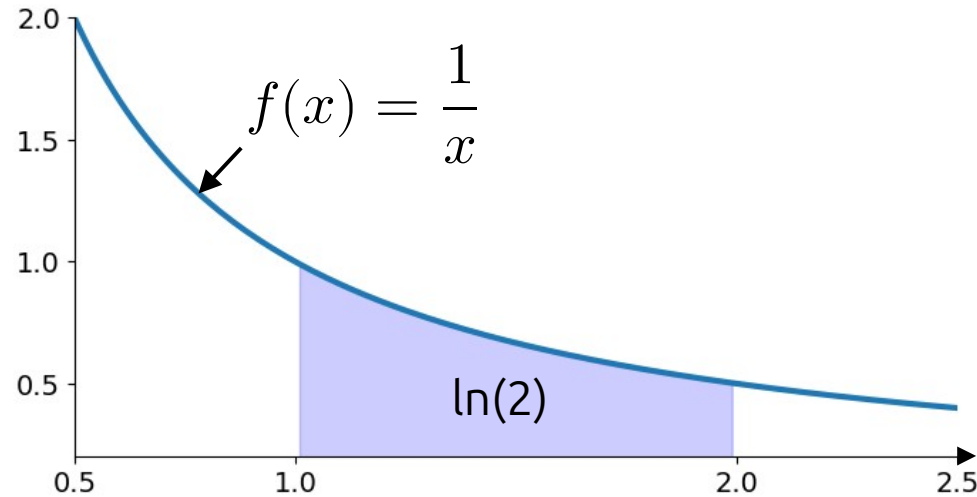
Numerical integration

All approximate $\int_a^b f(x) dx$

Name	Formula	Error bound
Trapezoidal rule	$\frac{h}{2}(f_0 + f_1)$	$\frac{M_2}{12}h^3$
Composite trapezoidal rule	$\frac{h}{2} \sum_{i=1}^n (f_{i-1} + f_i)$	$\frac{(b-a)M_2}{12}h^2$
Simpson's 1/3 rule	$\frac{h}{3}(f_0 + 4f_1 + f_2)$	$\frac{M_4}{90}h^5$
Composite 1/3 rule	$\frac{h}{3} \sum_{i=1}^{n/2} (f_{2i-2} + 4f_{2i-1} + f_{2i})$	$\frac{(b-a)M_4}{180}h^4$

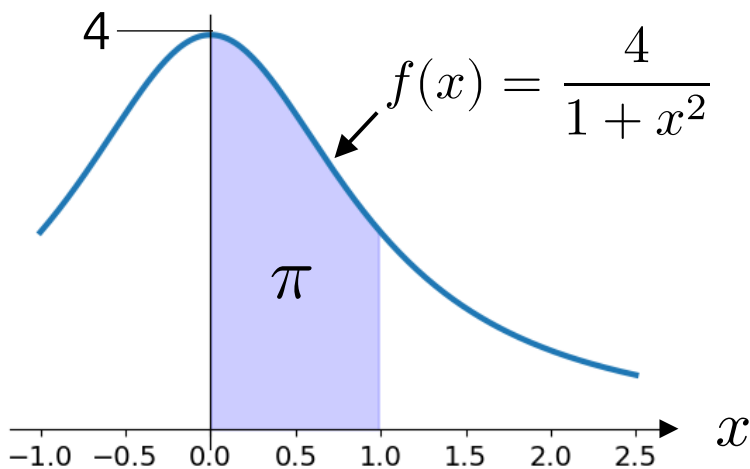
Example 1

Knowing that $\int_1^2 \frac{1}{x} dx = \ln(2)$, what is the minimum number of grid points needed to compute $\ln(2)$ with an error of at most 0.01, using either composite trapezoidal or composite Simpson?



Example 2

Knowing that $\int_0^1 \frac{1}{1+x^2} dx = \frac{\pi}{4}$, what is the minimum number of grid points needed to compute π with an error of at most 0.01, using either composite trapezoidal or composite Simpson?



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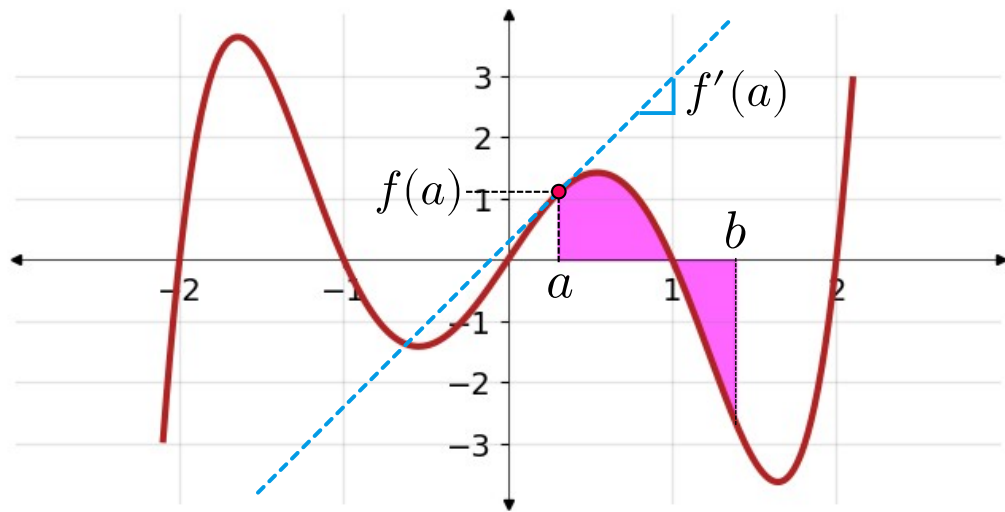
Lecture week 11-A

Numerical differentiation

Recall: Single variable calculus

- Derivative of $f(x)$:

$\frac{df}{dx}(a)$ or $f'(a)$ is the
slope of $f(x)$ at $x = a$

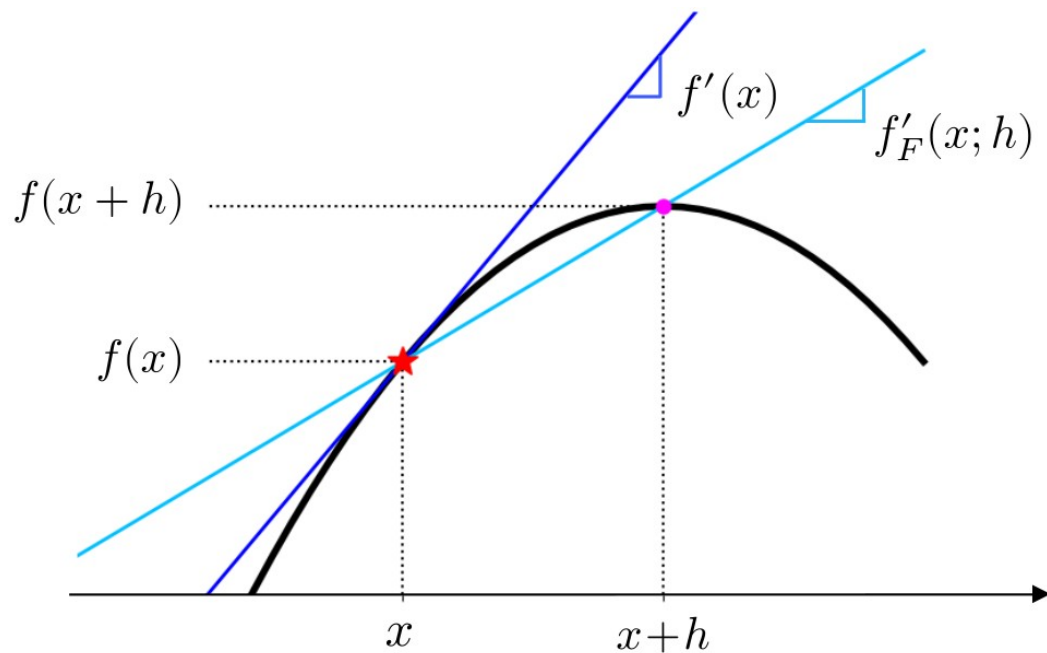


- Integral of $f(x)$ on the interval $[a, b]$:

$\int_a^b f(x) dx$ is the signed “area under the curve”
of $f(x)$ between a and b .

Approximating the derivative with finite differences

$$f'_F(x; h) = \frac{f(x + h) - f(x)}{h} = f'(x) + E_F(x; h)$$



Taylor's approximation of a function at a point

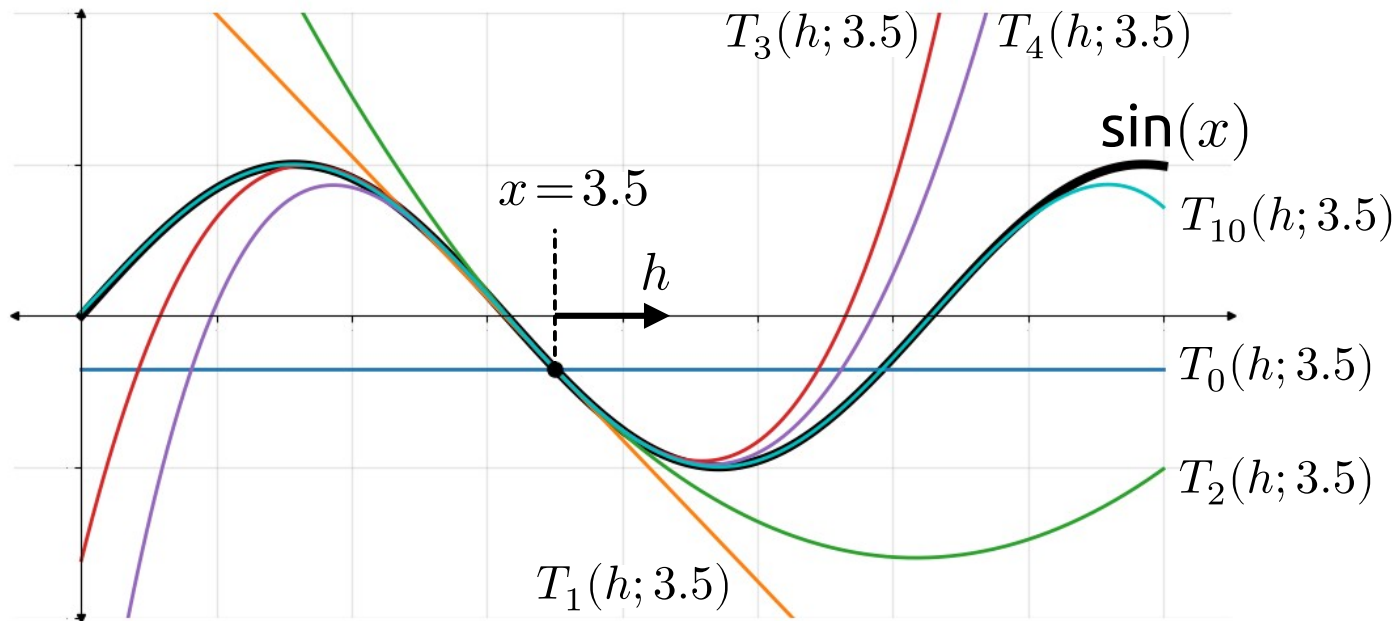
- $f : \mathbb{R} \rightarrow \mathbb{R}$ is a function with at least k derivatives.
- The k 'th order Taylor approximation of f about $x \in \mathbb{R}$ is a k 'th order polynomial in h , denoted $T_k(h; x)$.

$$\begin{aligned} f(x+h) &\approx T_k(h; x) = \sum_{i=0}^k \frac{f^{(i)}(x)}{i!} h^i \\ &= f(x) + f'(x)h + \frac{f''(x)}{2!}h^2 + \dots + \frac{f^{(k)}(x)}{k!}h^k \end{aligned}$$

- Note: $T_k(h; x)$ is a polynomial in h , with x held fixed.

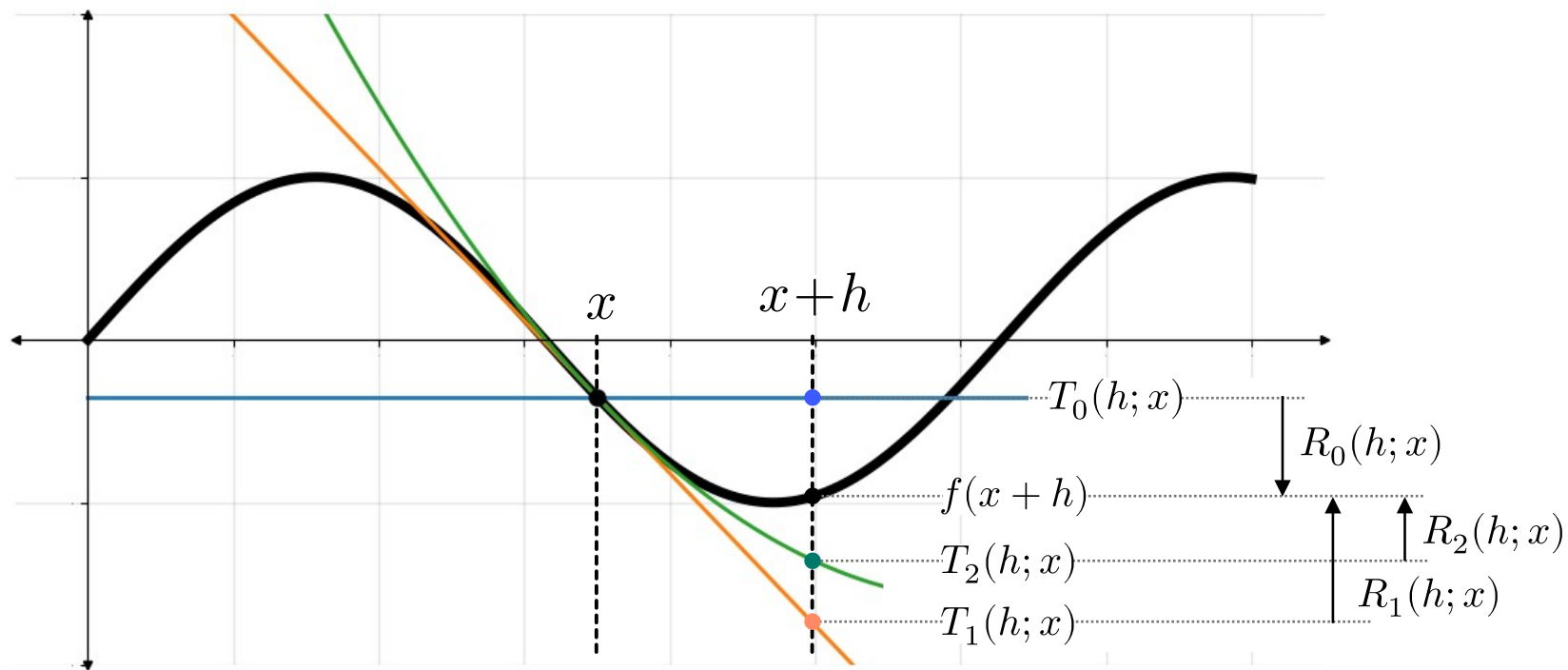
Example: Use a Taylor polynomial to approximate $\sin(x)$ near $x = 3.5$.

$$\sin(3.5+h) \approx \sin(3.5) + \cos(3.5)h - \frac{\sin(3.5)}{2!}h^2 - \frac{\cos(3.5)}{3!}h^3 + \dots$$



- $R_k(h; x)$ is the *approximation error* (a.k.a. the *residual*).

$$R_k(h; x) = f(x+h) - T_k(h; x)$$



Taylor's theorem

For any h and any k , there exists a ξ between x and $x+h$ such that

$$R_k(h; x) = \frac{f^{(k+1)}(\xi)}{(k+1)!} h^{k+1}$$

In particular,

$$|R_k(h; x)| \leq \frac{\max_{\xi \in [x, x+h]} |f^{(k+1)}(\xi)|}{(k+1)!} |h|^{k+1}$$

Bound the error of the forward differentiation scheme

- Apply Taylor's theorem to $f(x)$ with $k=1$, and $h>0$.

$$f(x+h) = f(x) + f'(x)h + \frac{f''(\xi)}{2}h^2 \quad \text{for some } \xi \in [x, x+h]$$

- Rearrange and take absolute value.

$$\left| \frac{f(x+h) - f(x)}{h} - f'(x) \right| = \frac{|f''(\xi)|}{2} h \quad \text{for some } \xi \in [x, x+h]$$

- The left hand side is the approximation error of $f'_F(x; h)$.

$$|E_F(x; h)| = \frac{|f''(\xi)|}{2} h \quad \text{for some } \xi \in [x, x+h]$$

- Since we don't know ξ , let' be conservative.

$$|E_F(x; h)| \leq \frac{\max_{\xi \in [x, x+h]} |f''(\xi)|}{2} h \leq \frac{M_2}{2} h \quad \dots O(h)$$

$$\text{where } M_2 = \max_{\xi \in \mathbb{R}} |f''(\xi)|$$

Other differencing schemes

- Backward difference: $f'_B(x; h) = \frac{f(x) - f(x - h)}{h}$

$$|E_B(h)| \leq \frac{M_2}{2} h \quad \dots O(h)$$

- Central difference: $f'_C(x; h) = \frac{f(x + h) - f(x - h)}{2h}$

$$|E_C(h)| \leq \frac{M_3}{6} h^2 \quad \dots O(h^2)$$

where $M_3 = \max_{\xi \in \mathbb{R}} |f'''(\xi)|$

Approximations for higher order derivatives

- Second derivative: $f_C''(x; h) = \frac{f(x + h) - 2f(x) + f(x - h)}{h^2}$

$$E_C(h) \leq \frac{M_4}{12} h^2$$

- In general, any $f^{(n)}(x)$ can be approximated using points around x .
- The more points, the better the estimate.
- See <https://web.media.mit.edu/~crtaylor/calculator.html>

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Lecture week 11-B Integration

A few applications of integration

- **Mechanics**

- Find centroids and moments of inertia of a body
- Find total forces acting on a body

- **Kinematics**

- Obtain velocities from accelerations and displacements from velocities

- **Signal processing**

- Calculate the energy of a signal

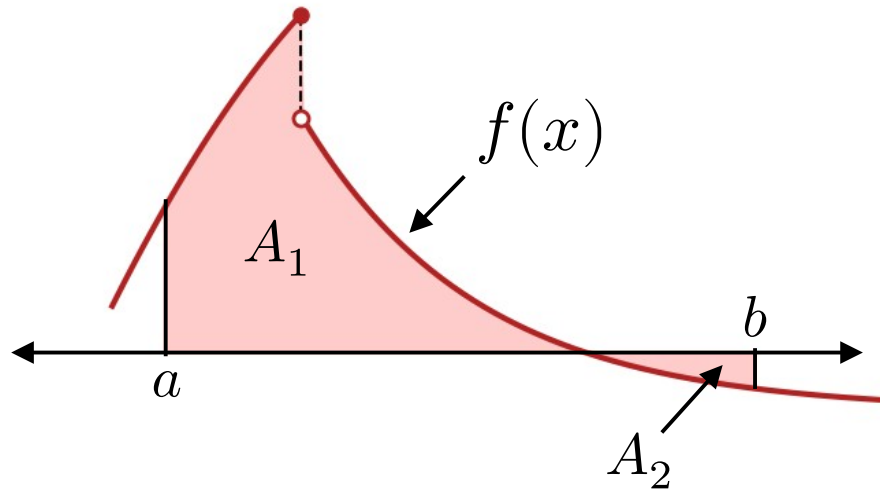
- **Hydrology**

- Find the total volume of a body of water

- ...

Definition of the integral

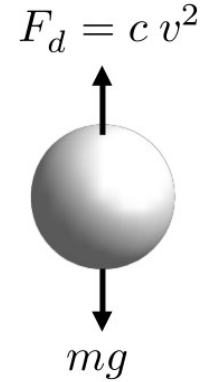
$\int_a^b f(x) dx$... signed area between $f(x)$ and the x axis.



Example: Find the instant at which a spherical object, dropped from a height of 15 meters, will hit the ground.

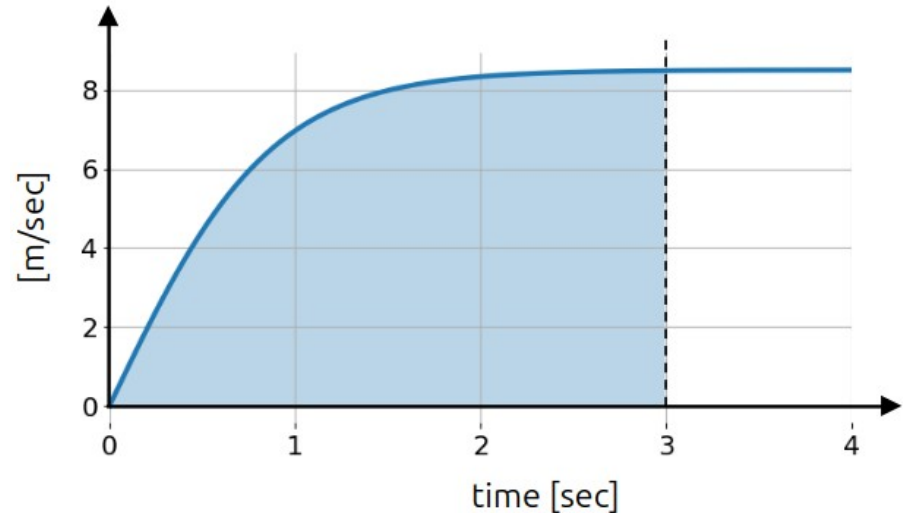
1) Apply Newton's second law:

$$mv'(t) = cv^2(t) - mg$$
$$v(0) = 0$$



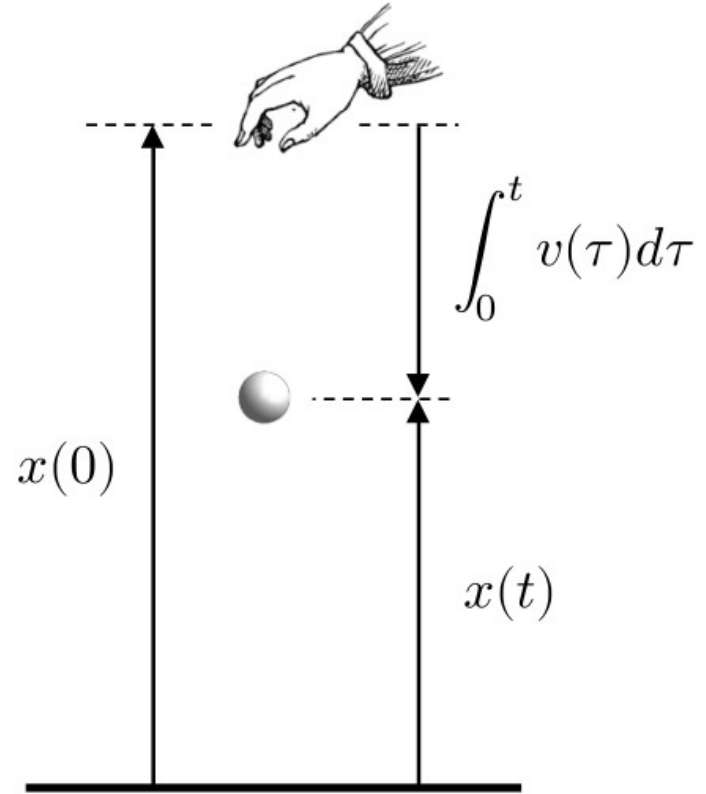
2) Solve ODE for the velocity:

$$v(t) = -\sqrt{\frac{gm}{c}} \tanh \left(\sqrt{\frac{gc}{m}} t \right)$$



3) Integrate the velocity to obtain position:

$$x(t) = x(0) + \int_0^t v(\tau) d\tau$$

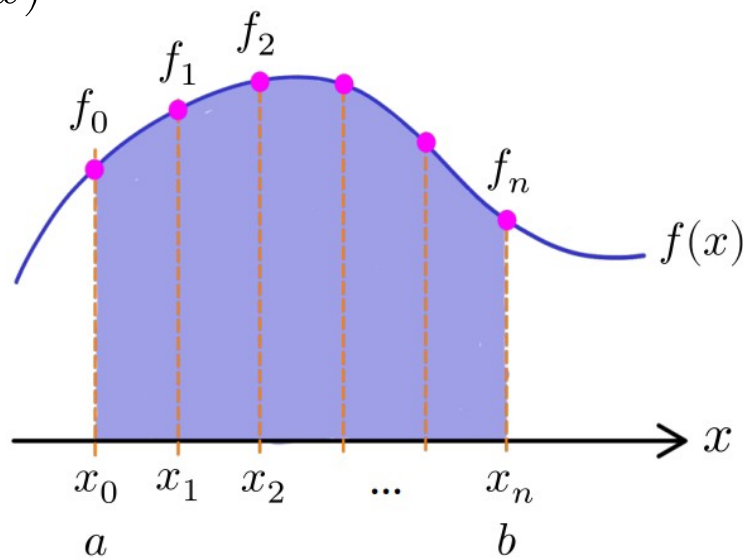


Newton-Cotes formulas

To evaluate $I = \int_a^b f(x) dx$

1. Split $[a, b]$ into n segments of width h
2. Approximate $f(x)$ using a polynomial $p(x)$
3. Integrate $p(x)$.

$$\tilde{I} = \int_a^b p(x) dx \approx I$$



Finding $p(x)$

- $p(x)$ is the simplest polynomial that goes through all $n + 1$ points.
- It is called the *Lagrange polynomial* and it is easy to compute.

$$\ell_i(x) = \frac{(x - x_0)}{(x_i - x_0)} \cdots \frac{(x - x_{i-1})}{(x_i - x_{i-1})} \frac{(x - x_{i+1})}{(x_i - x_{i+1})} \cdots \frac{(x - x_n)}{(x_i - x_n)} \quad \forall i \in \{0, \dots, n\}$$

$$p(x) = \sum_{i=0}^n f_i \ell_i(x)$$

Notice that:

$$1. \ell_i(x_i) = 1 \quad \forall i \in \{0, \dots, n\}$$

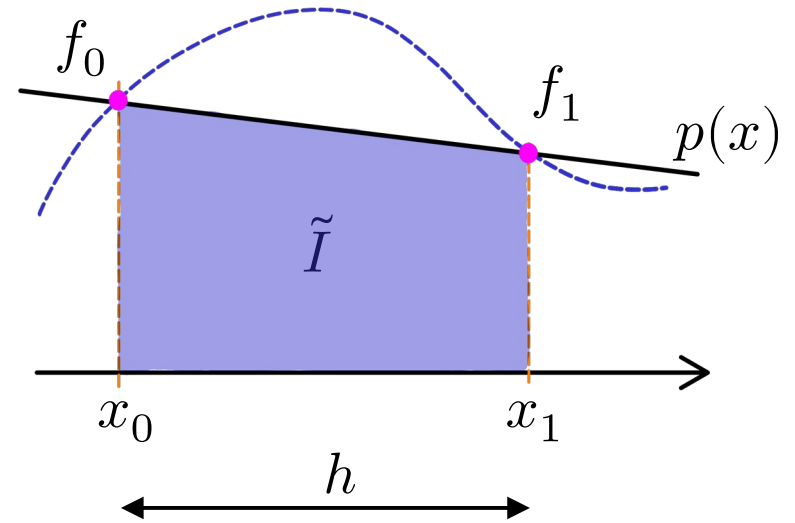
$$2. \ell_i(x_j) = 0 \quad \text{if } i \neq j$$

$$3. \text{order}(\ell(x)) = n \Rightarrow \text{order}(p(x)) \leq n$$

First order Newton-Cotes: The trapezoidal rule

$$p(x) = \frac{f_1 - f_0}{h} x + \frac{f_0 x_1 - f_1 x_0}{h}$$

$$\tilde{I} = \int_{x_0}^{x_1} p(x) dx = \frac{f_0 + f_1}{2} h$$



Error of the trapezoidal rule

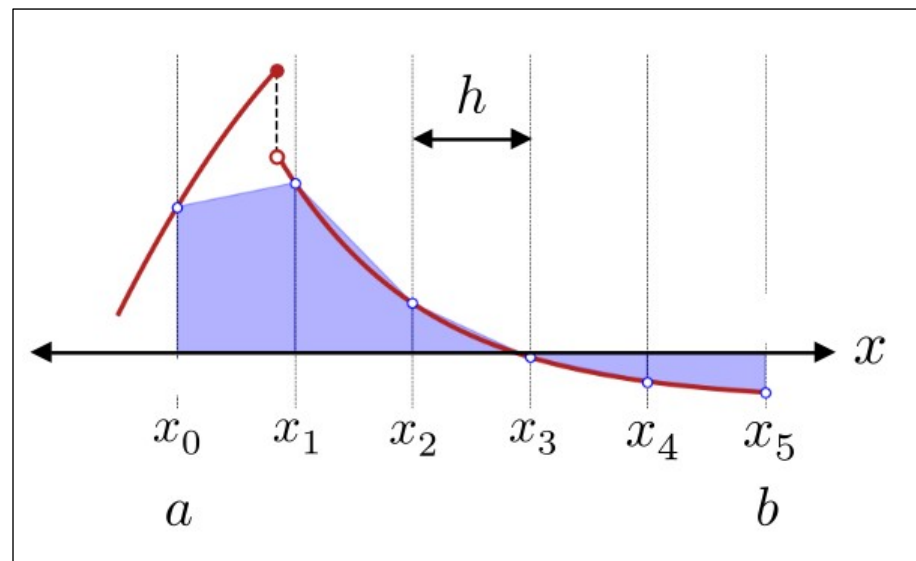
- If
1. f is continuous on the interval $[a, b]$
 2. f has bounded second derivative: $M_2 = \max_{x \in [a, b]} |f''(x)|$

Then
$$|I - \tilde{I}| \leq \frac{(b-a)^3}{12} M_2$$

Composite trapezoidal rule

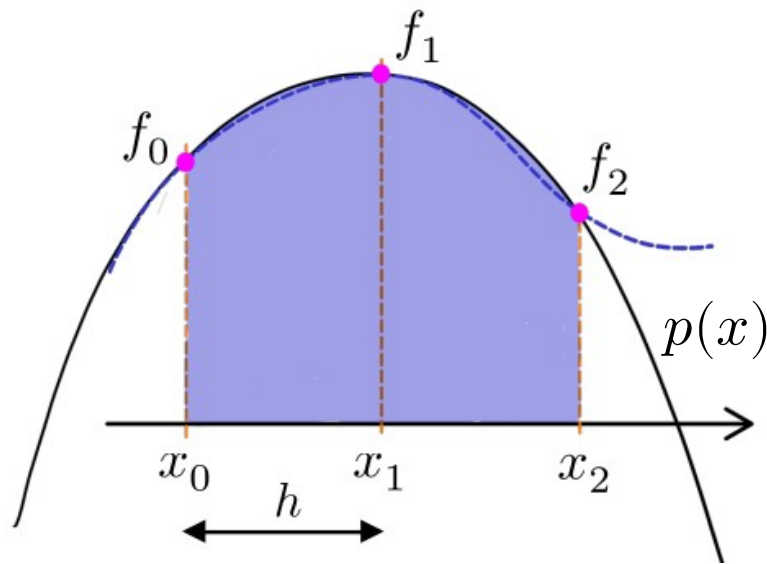
- Apply the trapezoidal rule to each segment.
- Accumulated error:

$$\begin{aligned} |\text{total error}| &= \sum_{i=1}^n |\text{error } i'\text{th segment}| \\ &\leq n \frac{h^3}{12} M_2 \\ &= \frac{(b-a)h^2}{12} M_2 \end{aligned}$$



Second order Newton-Cotes: Simpson's rule

$$p(x) = f_0 \frac{(x - x_1)}{(x_0 - x_1)} \frac{(x - x_2)}{(x_0 - x_2)} + f_1 \frac{(x - x_0)}{(x_1 - x_0)} \frac{(x - x_2)}{(x_1 - x_2)} + f_2 \frac{(x - x_0)}{(x_2 - x_0)} \frac{(x - x_1)}{(x_2 - x_1)}$$



without loss of generality, take $x_0 = -h$, $x_1 = 0$, $x_2 = h$

$$\begin{aligned} p(x) &= f_0 \frac{x}{h} \frac{(x-h)}{2h} + f_1 \frac{(x+h)}{h} \frac{(x-h)}{-h} + f_2 \frac{(x+h)}{2h} \frac{x}{h} \\ &= \vdots \\ &= \frac{1}{2h^2} \left((f_0 - 2f_1 + f_2) x^2 + (f_2 - f_0) h x + 2f_1 h^2 \right) \end{aligned}$$

$$\tilde{I} = \int_{-h}^h p(x) dx = \dots = \frac{h}{3} (f_0 + 4f_1 + f_2)$$

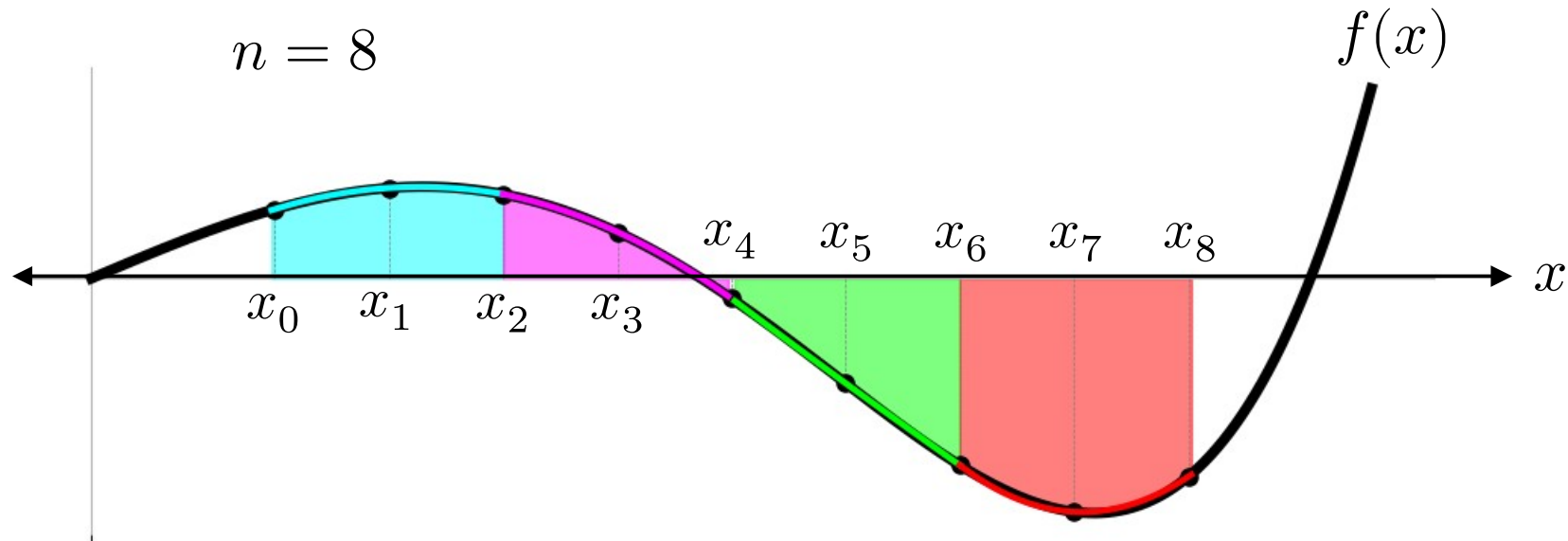
Error of Simpson's 1/3 rule

If 1. f is continuous on the interval $[a, b]$

2. f has bounded fourth derivative: $M_4 = \max_{x \in [a, b]} |f^{(4)}(x)|$

$$\text{Then } |I - \tilde{I}| \leq \frac{(b-a)^5}{2880} M_4$$

Composite 1/3 rule



$$|\text{total error}| \leq \frac{(b-a)h^4}{2800} M_4 \quad \dots O(h^4)$$

E7 Introduction to Computer Programming for Scientists and Engineers

Lecture week 12-A

Roots of functions

(a.k.a. Solutions of nonlinear equations)



Terminology

- An **equation** is an assertion that two expressions (the LHS and the RHS) evaluate to the same number.

e.g. $x^2 - 3x - 4 = 2x + 10$

- To **solve an equation** is to find the values of the unknowns that make it true.
- This is equivalent to finding the **roots (a.k.a. zeros) of a scalar function**.

Informal notation:

$$f(x) = x^2 - 5x - 14$$

Formal notation:

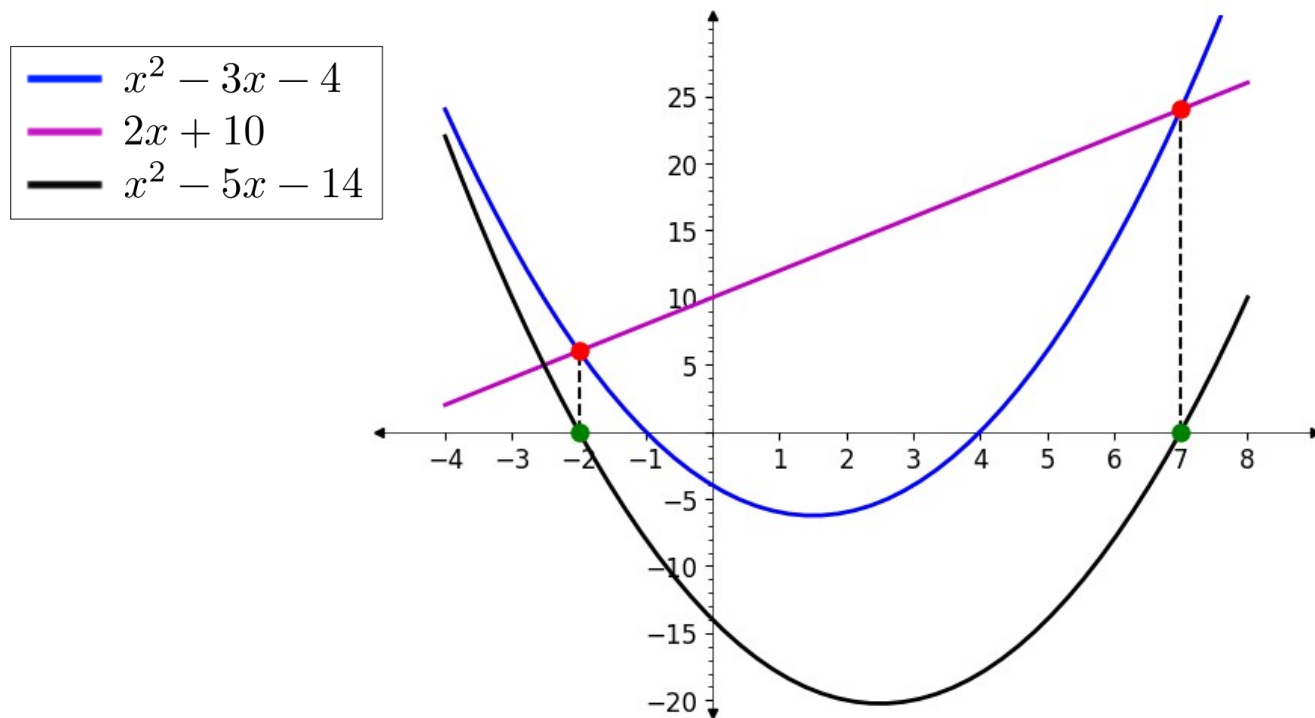
$$f : \mathbb{R} \rightarrow \mathbb{R}$$

$$x \mapsto x^2 - 5x - 14$$

Solutions of
 $x^2 - 3x - 4 = 2x + 10$

same as

Roots of
 $f(x) = x^2 - 5x - 14$



Terminology system/vector case

- An ***system of equations*** is an assertion that multiple equations hold simultaneously

e.g.
$$\begin{cases} x^2 - y = 1 \\ y^2 - x = 1 \end{cases}$$

- To ***solve a system of equations*** is to find the values of the unknowns that make all of its equations simultaneously true.
- This is equivalent to finding the ***roots (a.k.a. zeros) of a vector function***.

$$f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$$

$$(x, y) \mapsto (x^2 - y - 1, y^2 - x - 1)$$

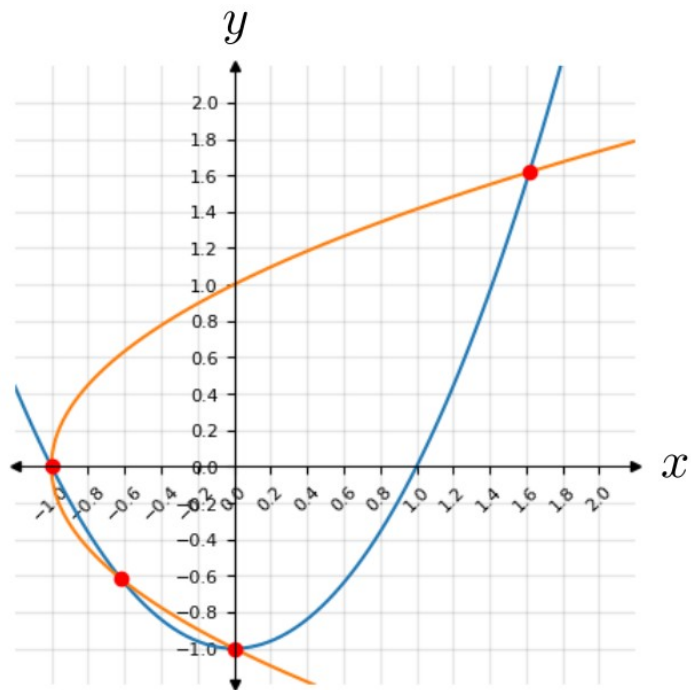
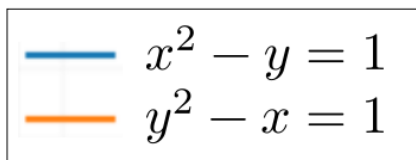
Solutions of

$$\begin{cases} x^2 - y = 1 \\ y^2 - x = 1 \end{cases}$$

same as

Roots of

$$f : (x, y) \mapsto (x^2 - y - 1, y^2 - x - 1)$$



Analytical formulas for the roots of polynomial functions

Quadratic: $f(x) = ax^2 + bx + c$

$$x_1 = \frac{b + \sqrt{b^2 - 4ac}}{2a}$$

$$x_2 = \frac{b - \sqrt{b^2 - 4ac}}{2a}$$

Cubic: $f(x) = ax^3 + bx^2 + cx + d$

$$\begin{aligned} x_1 &= -\frac{b}{3a} \\ &\quad - \frac{1}{3a} \sqrt[3]{\frac{1}{2} \left[2b^3 - 9abc + 27a^2d + \sqrt{(2b^3 - 9abc + 27a^2d)^2 - 4(b^2 - 3ac)^3} \right]} \\ &\quad - \frac{1}{3a} \sqrt[3]{\frac{1}{2} \left[2b^3 - 9abc + 27a^2d - \sqrt{(2b^3 - 9abc + 27a^2d)^2 - 4(b^2 - 3ac)^3} \right]} \\ x_2 &= -\frac{b}{3a} \\ &\quad + \frac{1 + i\sqrt{3}}{6a} \sqrt[3]{\frac{1}{2} \left[2b^3 - 9abc + 27a^2d + \sqrt{(2b^3 - 9abc + 27a^2d)^2 - 4(b^2 - 3ac)^3} \right]} \\ &\quad + \frac{1 - i\sqrt{3}}{6a} \sqrt[3]{\frac{1}{2} \left[2b^3 - 9abc + 27a^2d - \sqrt{(2b^3 - 9abc + 27a^2d)^2 - 4(b^2 - 3ac)^3} \right]} \\ x_3 &= -\frac{b}{3a} \\ &\quad + \frac{1 - i\sqrt{3}}{6a} \sqrt[3]{\frac{1}{2} \left[2b^3 - 9abc + 27a^2d + \sqrt{(2b^3 - 9abc + 27a^2d)^2 - 4(b^2 - 3ac)^3} \right]} \\ &\quad + \frac{1 + i\sqrt{3}}{6a} \sqrt[3]{\frac{1}{2} \left[2b^3 - 9abc + 27a^2d - \sqrt{(2b^3 - 9abc + 27a^2d)^2 - 4(b^2 - 3ac)^3} \right]} \end{aligned}$$

Quartic: $f(x) = ax^4 + bx^3 + cx^2 + dx + e$

$$\begin{aligned} r_1 &= \frac{1}{4} \sqrt{\frac{1}{3} \left(\frac{2^{\frac{1}{2}} (b^3 - 3ac + 12d)}{3 \left(2b^3 - 9abc + 27c^2 + 27ad - 72bd + \sqrt{-4b^3 - 3ac + 12d} \right) + \left(2b^3 - 9abc + 27c^2 + 27ad - 72bd + \sqrt{-4b^3 - 3ac + 12d} \right)^2} \right)^{\frac{1}{2}}} - \frac{1}{4} \sqrt{\frac{1}{3} \left(\frac{2^{\frac{1}{2}} (b^3 - 3ac + 12d)}{3 \left(2b^3 - 9abc + 27c^2 + 27ad - 72bd + \sqrt{-4b^3 - 3ac + 12d} \right) + \left(2b^3 - 9abc + 27c^2 + 27ad - 72bd + \sqrt{-4b^3 - 3ac + 12d} \right)^2} \right)^{\frac{1}{2}}} \\ r_2 &= \frac{1}{4} \sqrt{\frac{1}{3} \left(\frac{2^{\frac{1}{2}} (b^3 - 3ac + 12d)}{3 \left(2b^3 - 9abc + 27c^2 + 27ad - 72bd + \sqrt{-4b^3 - 3ac + 12d} \right) + \left(2b^3 - 9abc + 27c^2 + 27ad - 72bd + \sqrt{-4b^3 - 3ac + 12d} \right)^2} \right)^{\frac{1}{2}}} + \frac{1}{4} \sqrt{\frac{1}{3} \left(\frac{2^{\frac{1}{2}} (b^3 - 3ac + 12d)}{3 \left(2b^3 - 9abc + 27c^2 + 27ad - 72bd + \sqrt{-4b^3 - 3ac + 12d} \right) + \left(2b^3 - 9abc + 27c^2 + 27ad - 72bd + \sqrt{-4b^3 - 3ac + 12d} \right)^2} \right)^{\frac{1}{2}}} \\ r_3 &= \frac{1}{4} \sqrt{\frac{1}{3} \left(\frac{2^{\frac{1}{2}} (b^3 - 3ac + 12d)}{3 \left(2b^3 - 9abc + 27c^2 + 27ad - 72bd + \sqrt{-4b^3 - 3ac + 12d} \right) + \left(2b^3 - 9abc + 27c^2 + 27ad - 72bd + \sqrt{-4b^3 - 3ac + 12d} \right)^2} \right)^{\frac{1}{2}}} - \frac{1}{4} \sqrt{\frac{1}{3} \left(\frac{2^{\frac{1}{2}} (b^3 - 3ac + 12d)}{3 \left(2b^3 - 9abc + 27c^2 + 27ad - 72bd + \sqrt{-4b^3 - 3ac + 12d} \right) + \left(2b^3 - 9abc + 27c^2 + 27ad - 72bd + \sqrt{-4b^3 - 3ac + 12d} \right)^2} \right)^{\frac{1}{2}}} \\ r_4 &= \frac{1}{4} \sqrt{\frac{1}{3} \left(\frac{2^{\frac{1}{2}} (b^3 - 3ac + 12d)}{3 \left(2b^3 - 9abc + 27c^2 + 27ad - 72bd + \sqrt{-4b^3 - 3ac + 12d} \right) + \left(2b^3 - 9abc + 27c^2 + 27ad - 72bd + \sqrt{-4b^3 - 3ac + 12d} \right)^2} \right)^{\frac{1}{2}}} + \frac{1}{4} \sqrt{\frac{1}{3} \left(\frac{2^{\frac{1}{2}} (b^3 - 3ac + 12d)}{3 \left(2b^3 - 9abc + 27c^2 + 27ad - 72bd + \sqrt{-4b^3 - 3ac + 12d} \right) + \left(2b^3 - 9abc + 27c^2 + 27ad - 72bd + \sqrt{-4b^3 - 3ac + 12d} \right)^2} \right)^{\frac{1}{2}}} \end{aligned}$$

Quintic: $f(x) = ax^5 + bx^4 + cx^3 + dx^2 + ex + g$

Abel–Ruffini theorem (1824) : No formula exists!

Numerical methods

We will study three numerical methods (a.k.a. algorithms) for finding the roots of functions.

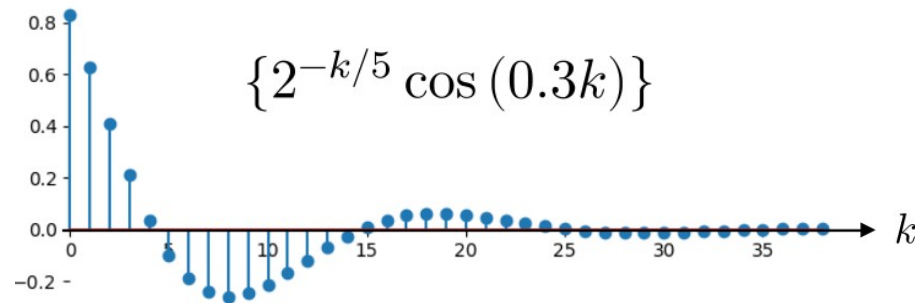
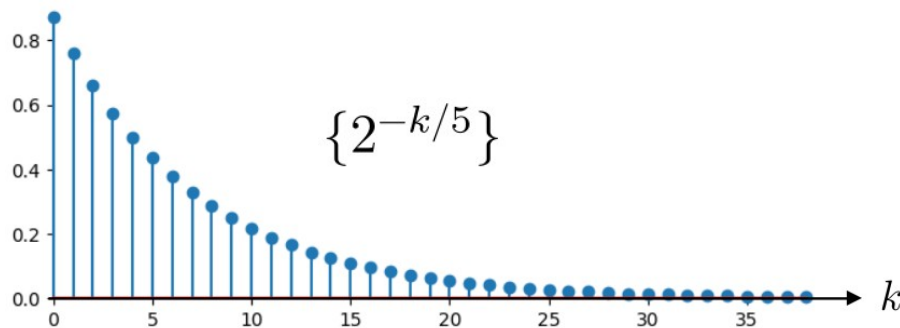
- 1) Bisection
- 2) Newton-Raphson
- 3) Fixed point iteration

Each of these algorithms produces a sequence of numbers that (hopefully) converges to a root.

Sequences and convergence

- A **sequence** is an infinite list of numbers or vectors. $\{x_0, x_1, x_2, \dots\}$
- Notation: $\{x_k\}$
- A **convergent sequence** is a sequence that tends to a fixed value.

Example: Both of these sequences converge to 0.



Bisection

- Requirements: f is $\mathbb{R} \rightarrow \mathbb{R}$ and continuous.

Choose two points N_0 and P_0 such that
 $f(N_0) < 0$ and $f(P_0) > 0$

$k = 0$

repeat until convergence:

$$C = (N_k + P_k)/2$$

if $f(C) < 0$

$$N_{k+1} = C$$

$$P_{k+1} = P_k$$

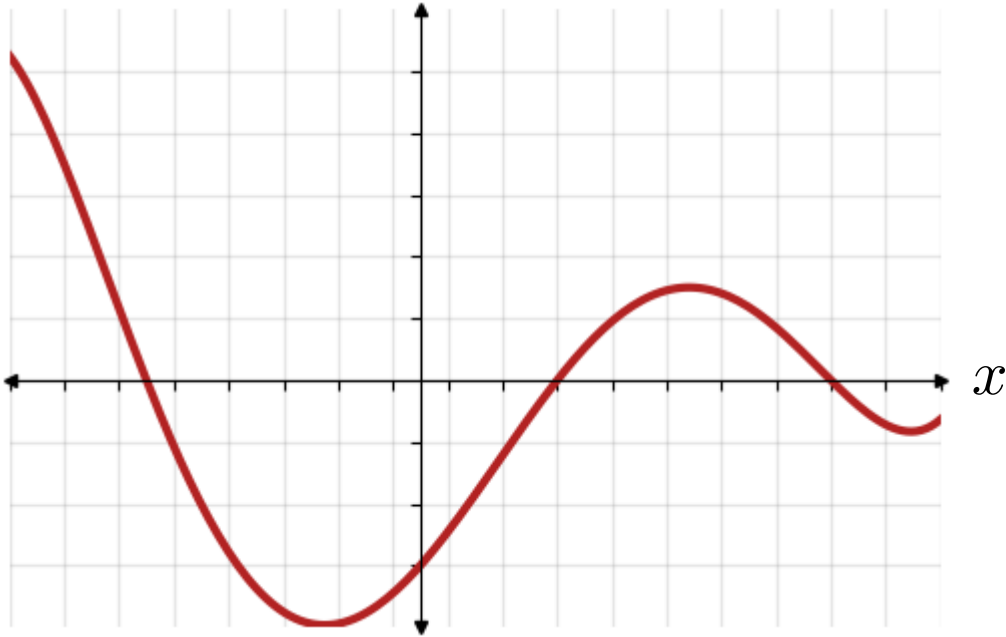
else

$$P_{k+1} = C$$

$$N_{k+1} = N_k$$

$$k = k + 1$$

Example: Bisection



Number of steps to achieve a desired tolerance

- Interval width after k steps

$$|P_k - N_k| = \frac{|P_0 - N_0|}{2^k}$$

- Steps to achieve width smaller than a given ϵ .

Newton-Raphson

- Requirements: f is $\mathbb{R} \rightarrow \mathbb{R}$ and continuous and differentiable.

x_0 is an initial guess

$k = 0$

repeat

$$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}$$

$k = k + 1$

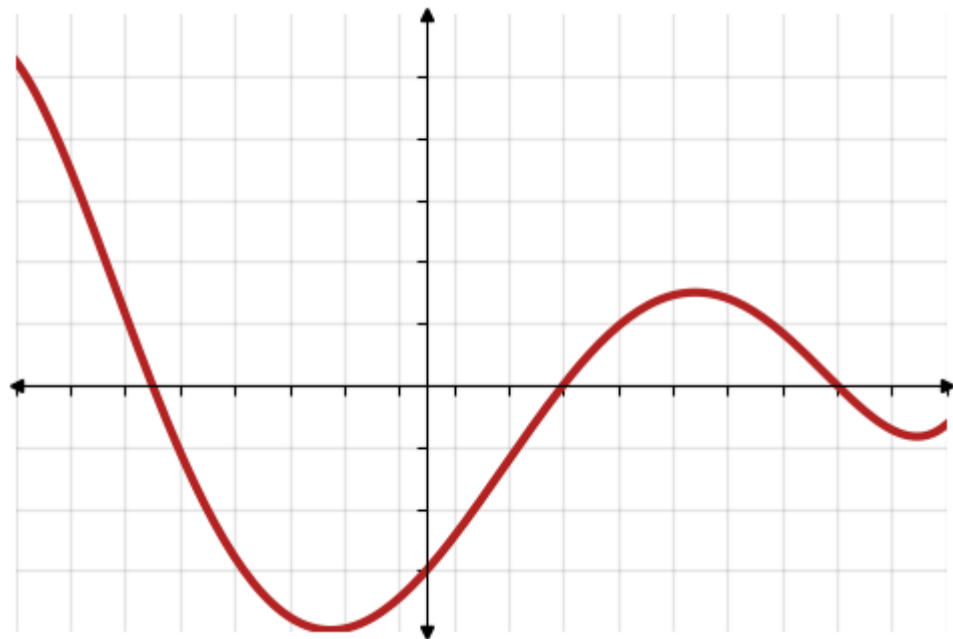
until

$$|x_{k+1} - x_k| < \epsilon_x$$

$$\text{or } |f(x_k)| < \epsilon_f$$

or $k > \text{maxsteps}$

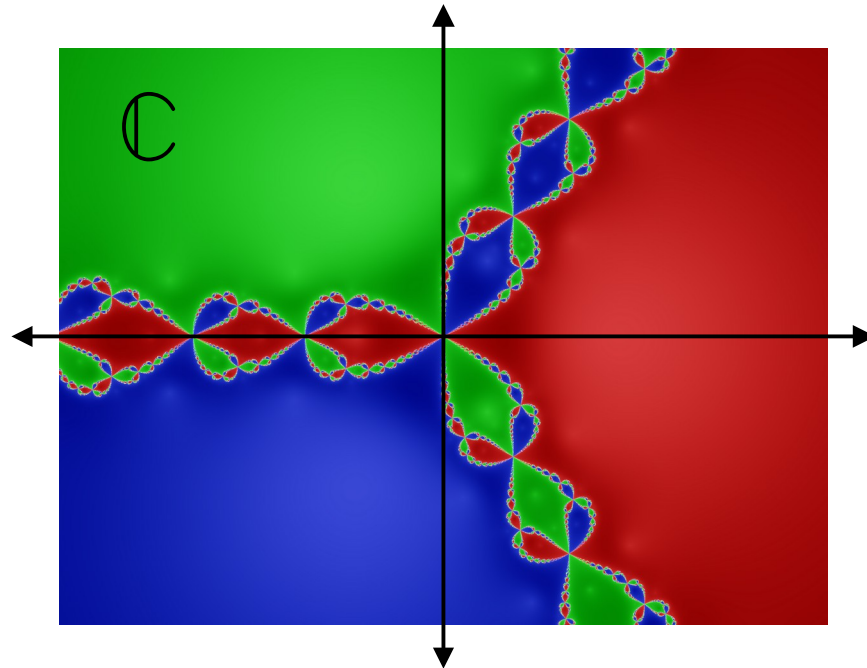
Example: Newton Raphson



Newton-Raphson makes fractals!

$$f(x) : \mathbb{C} \rightarrow \mathbb{C}$$

$$f(x) = x^3 - 1$$



Summary

	Bisection	Newton-Raphson
Requirements	Continuous scalar function.	Continuous function, with derivative.
Initialization	N_0, P_0 with $f(N_0) < 0, f(P_0) > 0$	any x_0
Failure cases	Function has no roots	Function has no roots, Derivative becomes 0

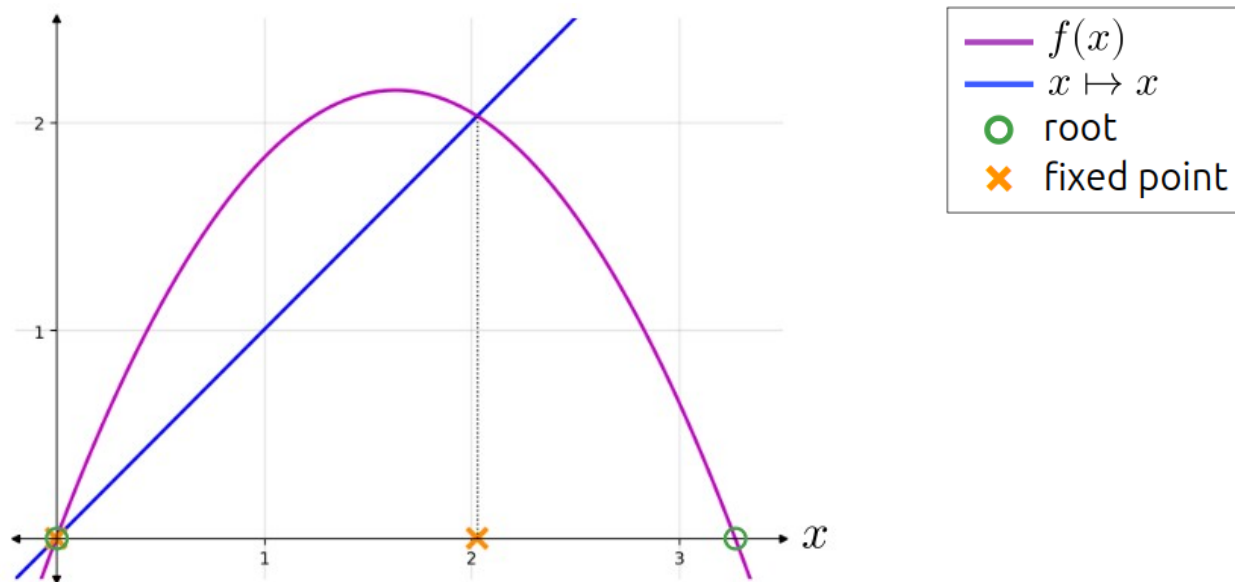
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Lecture week 12-B

Fixed points of functions

Terminology

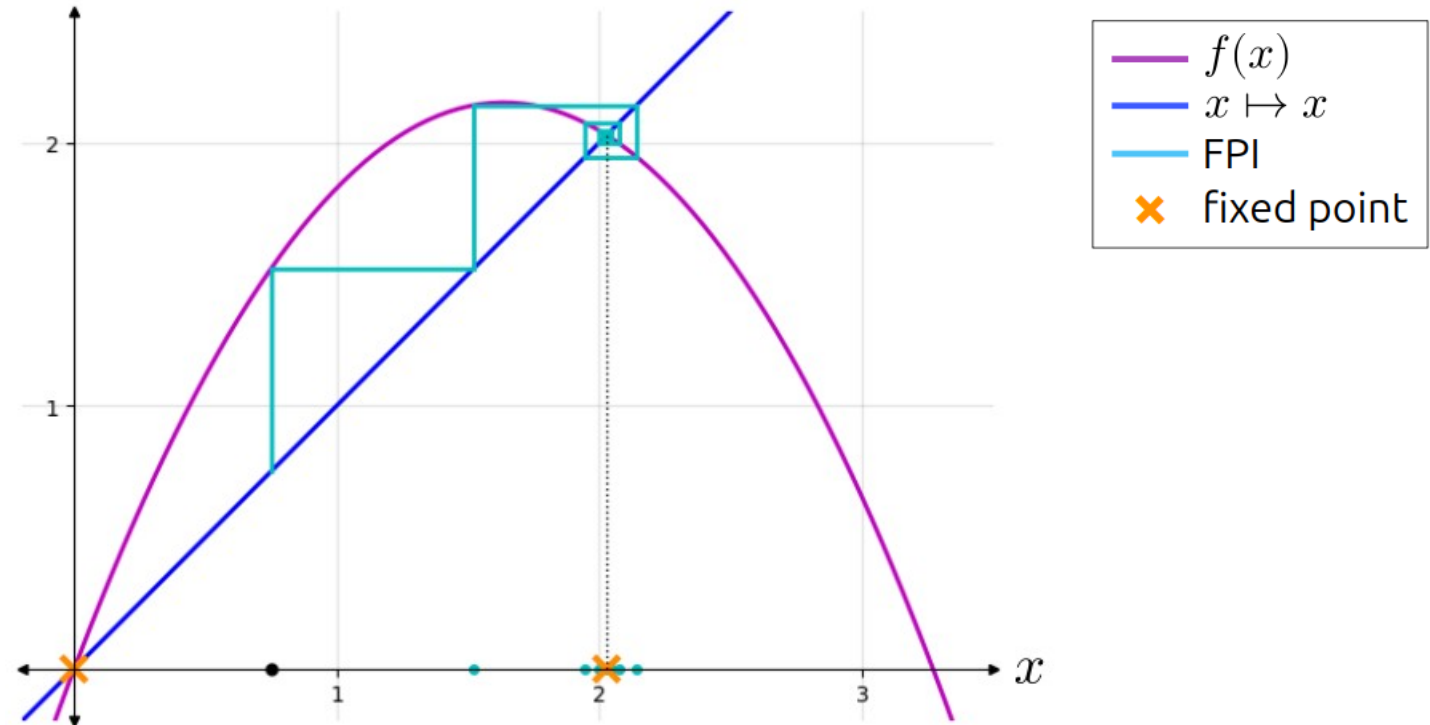
- A **fixed point** of a function $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is a point that maps to itself.
- That is, x^* is a fixed point of f if $f(x^*) = x^*$



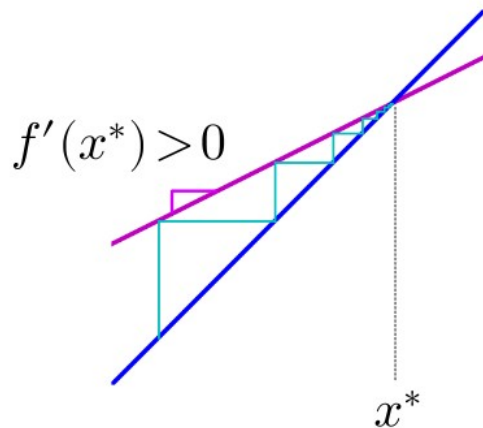
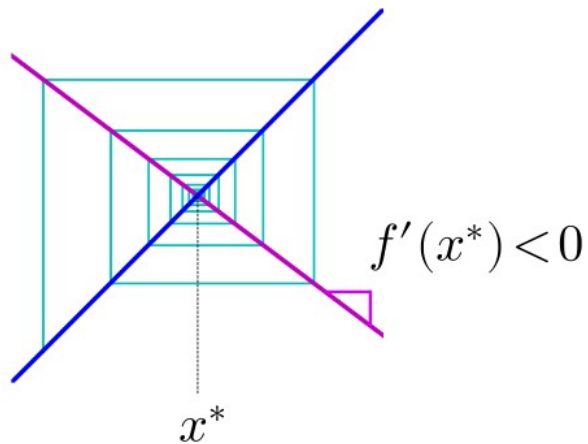
- A ***fixed point iteration*** (FPI) of is a sequence generated with,

choose x_0

$$x_{k+1} = f(x_k)$$



Convergence of FPI: Scalar functions



- linear case: $f(x) = ax + b$

FPI converges to $x^* = b/(1 - a)$ for any x_0 if and only if $|a| < 1$.

- nonlinear case: $f(x) : \mathbb{R} \rightarrow \mathbb{R}$

FPI converges to $x^* \in [a, b]$ for any $x_0 \in [a, b]$ if and only if $\max_{x \in [a, b]} |f'(x)| < 1$.

Using FPI to solving nonlinear equations

- We wish to find a root of $f(x)$.
- Apply FPI to a function $g(x)$ whose fixed points are roots of $f(x)$

$$g(x^*) = x^* \quad \Rightarrow \quad f(x^*) = 0$$

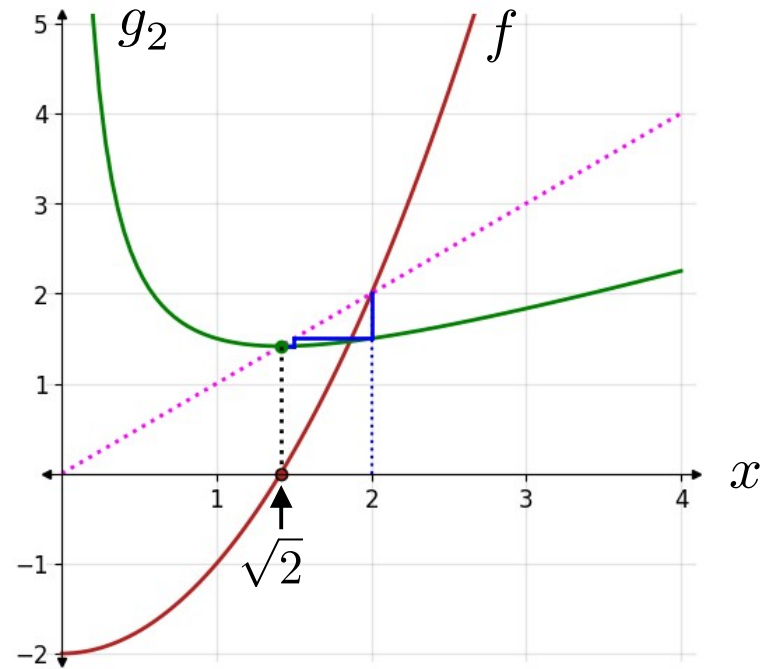
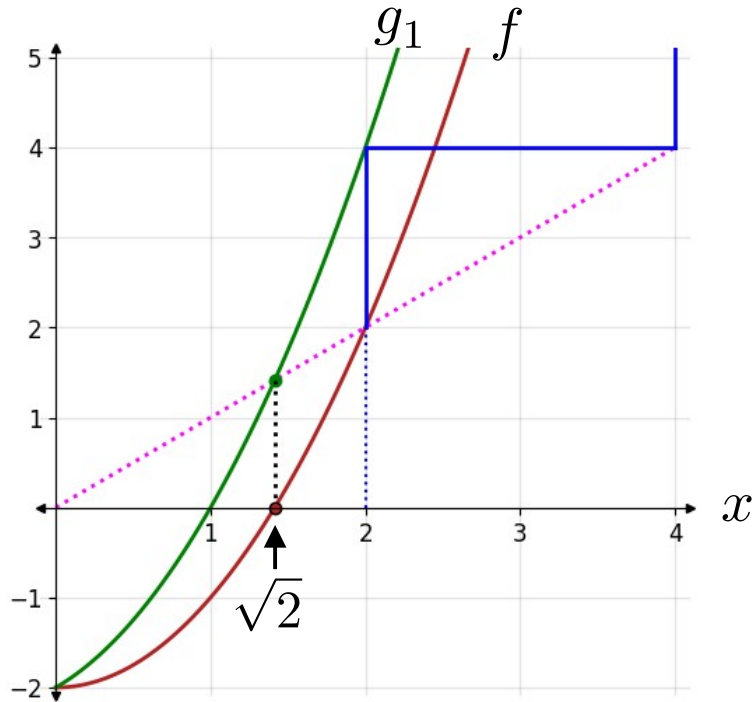
- How to choose $g(x)$?
- $g(x) = f(x) + x$ always works, but there may be better options.
- The shallower, the better.

Example: Find the positive root of $f(x) = x^2 - a$

$$g_1(x) = f(x) + x$$

$$g_2(x) = \frac{1}{2} \left(\frac{a}{x} + x \right)$$

e.g. $a = 2$



Using FPI to solve systems of nonlinear equations

- f is a system of n nonlinear equations in n unknowns. $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$
- Suppose R is some given region in $\mathbb{R}^n$
- Theorem:

If f has a fixed point in R , then FPI will converge for any $x_0 \in R$ provided all partial derivatives of f have absolute value less than 1 everywhere in R .

- This is a *conservative* statement, meaning that FPI *may* converge even when some of the partial derivatives are larger than 1.

Example: Solve
$$\begin{cases} x^2 - y = 1 \\ y^2 - x = 1 \end{cases}$$

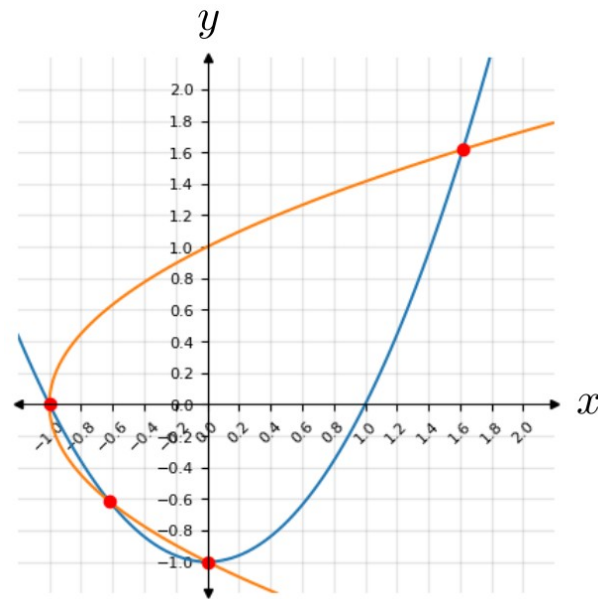
- Find the roots of

$$f : (x, y) \mapsto (x^2 - y - 1, y^2 - x - 1)$$

- Two options for FPI:

$$g : \begin{cases} x = y^2 - 1 \\ y = x^2 - 1 \end{cases} \Rightarrow \begin{aligned} g_1(x, y) &= y^2 - 1 \\ g_2(x, y) &= x^2 - 1 \end{aligned}$$

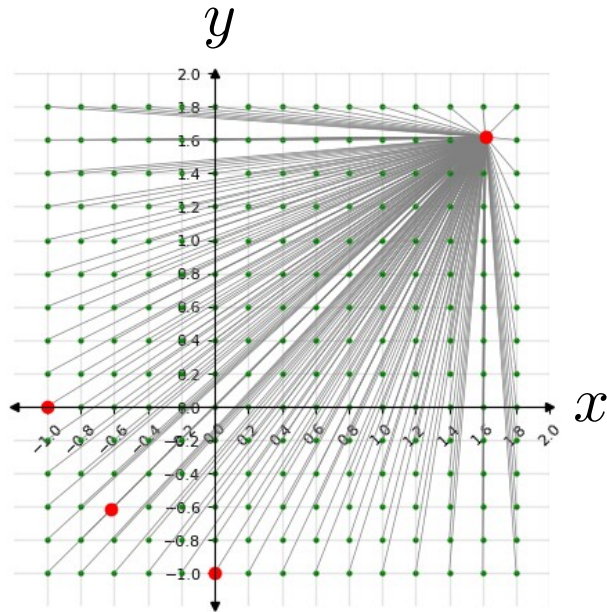
$$h : \begin{cases} x = \sqrt{1 + y} \\ y = \sqrt{1 + x} \end{cases} \Rightarrow \begin{aligned} h_1(x, y) &= \sqrt{1 + y} \\ h_2(x, y) &= \sqrt{1 + x} \end{aligned}$$



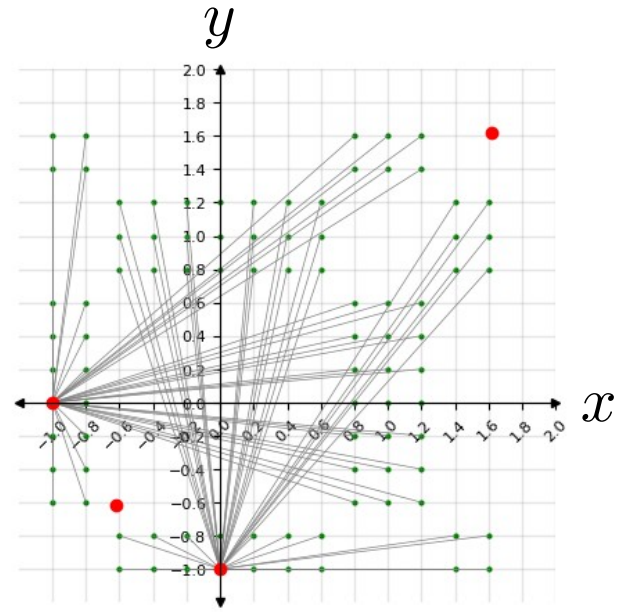
- Convergence analysis

- Perform FPI over a grid of initial conditions.
- Green dots are initial conditions for which FPI converged.
- Gray lines point to the fixed points to which it converged.

FPI on g_1



FPI on g_2



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Discussion week 12

Solving systems of nonlinear equations with FPI

Example: Find a solution to the following system of equations.

$$\begin{cases} 10\xi_1 + \xi_2 = 1 - \mathbf{\sin}(\xi_1 + \xi_2) \\ -\xi_1 + 10\xi_2 = 2 + \mathbf{\cos}(\xi_1 - \xi_2) \end{cases}$$

Candidate fixed point function #1.

$$\begin{cases} \xi_1 = 10\xi_2 - 2 - \mathbf{cos}(\xi_1 - \xi_2) \\ \xi_2 = -10\xi_1 + 1 - \mathbf{sin}(\xi_1 + \xi_2) \end{cases}$$

Candidate fixed point function #2. g_A

$$\begin{cases} \xi_1 = 0.1(1 - \xi_2 - \mathbf{sin}(\xi_1 + \xi_2)) \\ \xi_2 = 0.1(2 + \xi_1 + \mathbf{cos}(\xi_1 - \xi_2)) \end{cases}$$

Candidate fixed point function #3. g_B

- Change of variables.

$$\begin{array}{lcl} \eta_1 = \xi_1 + \xi_2 & & \xi_1 = (\eta_1 + \eta_2)/2 \\ \eta_2 = \xi_1 - \xi_2 & \Longleftrightarrow & \xi_2 = (\eta_1 - \eta_2)/2 \end{array}$$

- g_B

$$\begin{cases} \eta_1 = (1/11) \left(-9\eta_2 + 2 - 2 \sin(\eta_1) \right) \\ \eta_2 = (1/11) \left(9\eta_1 - 4 - 2 \cos(\eta_2) \right) \end{cases}$$

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Lecture week 13-A

Linear spatial transformations

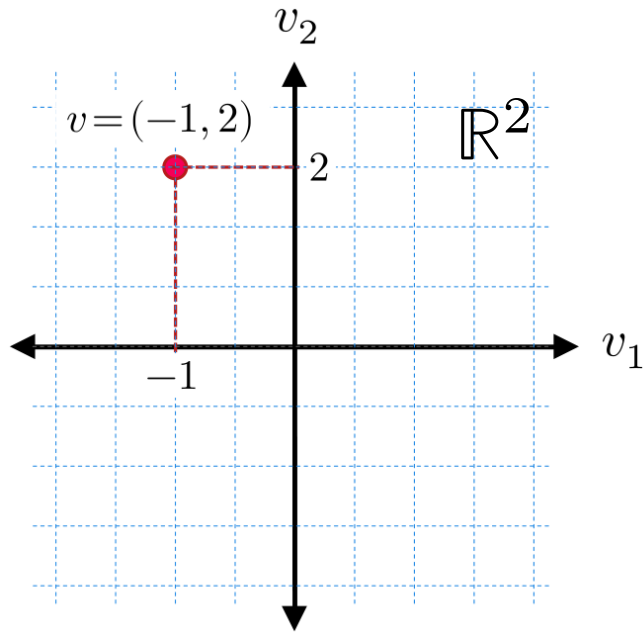
Matrix multiplication in NumPy

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix} =$$

see lecture notebook for code...

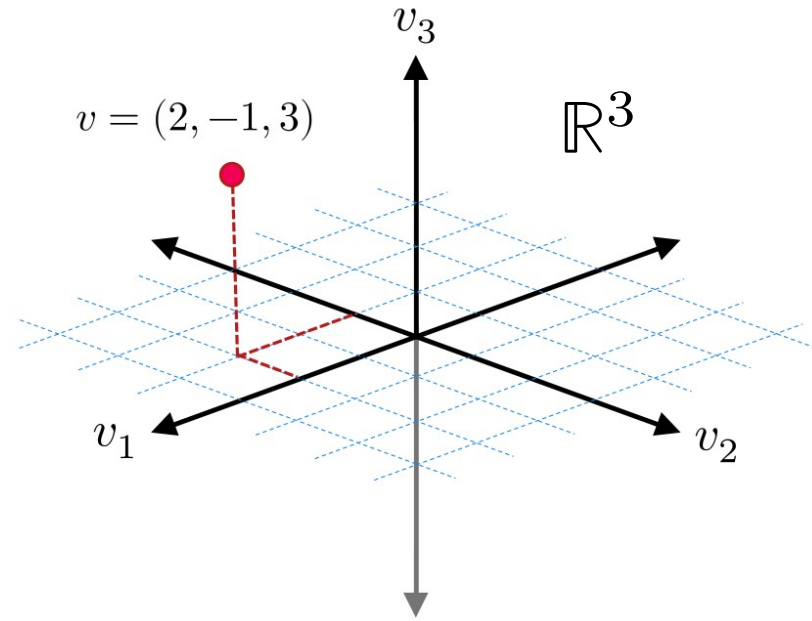
2D vector space

$$v = (v_1, v_2) \in \mathbb{R}^2$$



3D vector space

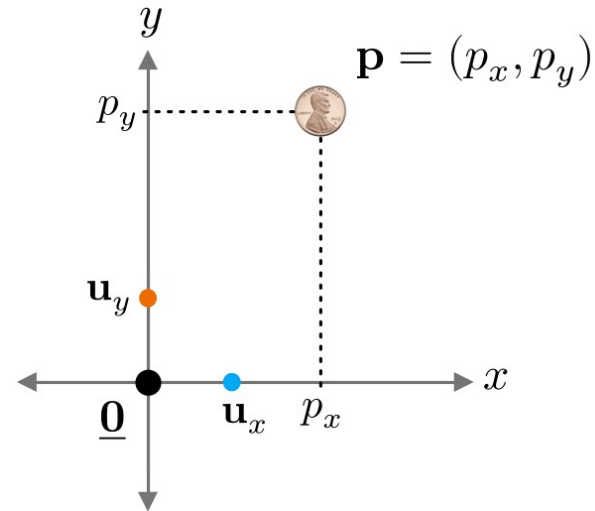
$$v = (v_1, v_2, v_3) \in \mathbb{R}^3$$



Cartesian coordinate system

- A Cartesian coordinate system is a tool for describing the positions of objects in space.
- It consists of several landmarks: the **origin** and several **unit vectors**.
- The line segments between the origin and each of the unit vectors must form **right angles**.

- $\underline{0} = (0, 0)$... origin
- $\mathbf{u}_x = (1, 0)$... unit vector
- $\mathbf{u}_y = (0, 1)$... unit vector



Coordinate transformation

A **coordinate transformation** is a function that maps the representation of an object in one coordinate system to its representation in another.

Example: 2D rotation

- Two coordinate systems with the same origin.

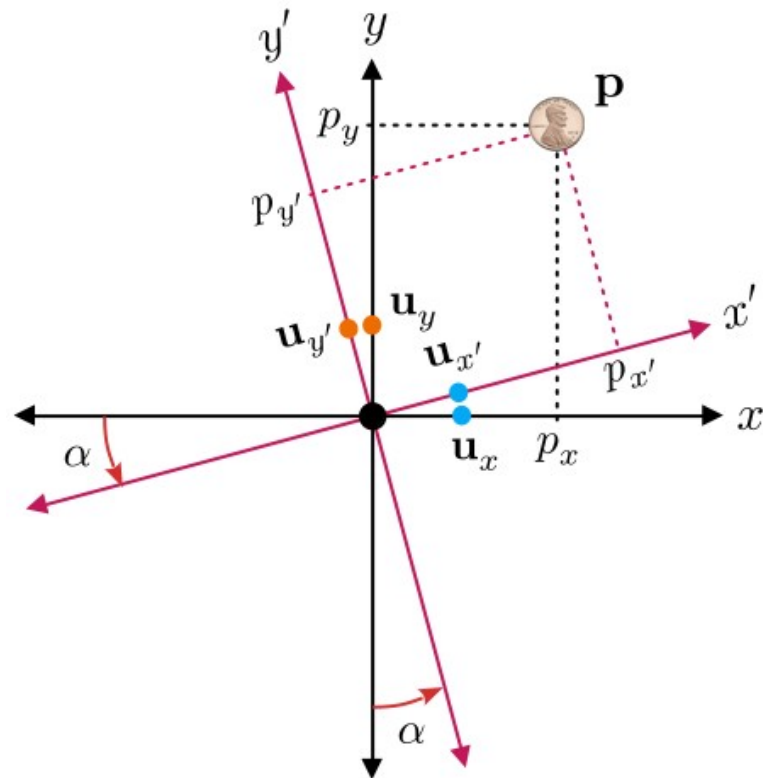
$$C_1 = (\underline{0}, \mathbf{u}_x, \mathbf{u}_y)$$

$$C_2 = (\underline{0}, \mathbf{u}_{x'}, \mathbf{u}_{y'})$$

- The same point has different representations in C_1 and C_2 .

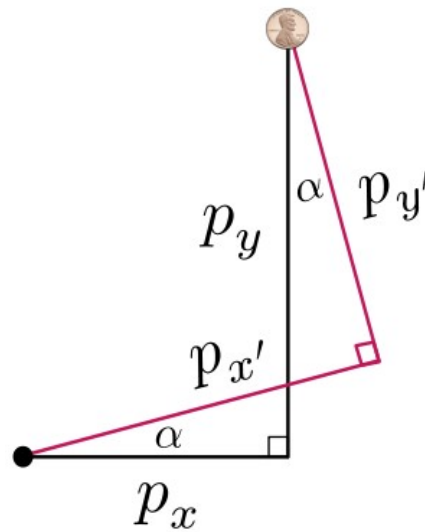
$$\mathbf{p} = (p_x, p_y) \quad \dots C_1$$

$$\mathbf{p} = (p_{x'}, p_{y'}) \quad \dots C_2$$



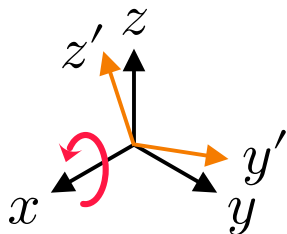
- Coordinate transformation: $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$
 $(p_x, p_y) \mapsto (p_{x'}, p_{y'})$
- T is a *linear transformation*, i.e. a *matrix multiplication*.

$$\begin{bmatrix} p_{x'} \\ p_{y'} \end{bmatrix} = \begin{bmatrix} \cos(\alpha) & \sin(\alpha) \\ -\sin(\alpha) & \cos(\alpha) \end{bmatrix} \begin{bmatrix} p_x \\ p_y \end{bmatrix}$$



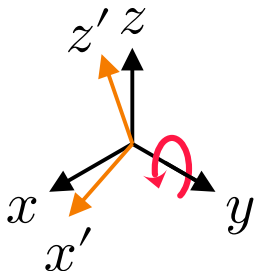
3D axis-aligned rotations

x axis:



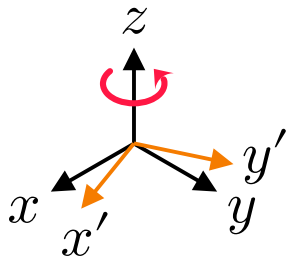
$$\begin{bmatrix} p_{x'} \\ p_{y'} \\ p_{z'} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(\alpha) & \sin(\alpha) \\ 0 & -\sin(\alpha) & \cos(\alpha) \end{bmatrix} \begin{bmatrix} p_x \\ p_y \\ p_z \end{bmatrix}$$

y axis:



$$\begin{bmatrix} p_{x'} \\ p_{y'} \\ p_{z'} \end{bmatrix} = \begin{bmatrix} \cos(\alpha) & 0 & -\sin(\alpha) \\ 0 & 1 & 0 \\ \sin(\alpha) & 0 & \cos(\alpha) \end{bmatrix} \begin{bmatrix} p_x \\ p_y \\ p_z \end{bmatrix}$$

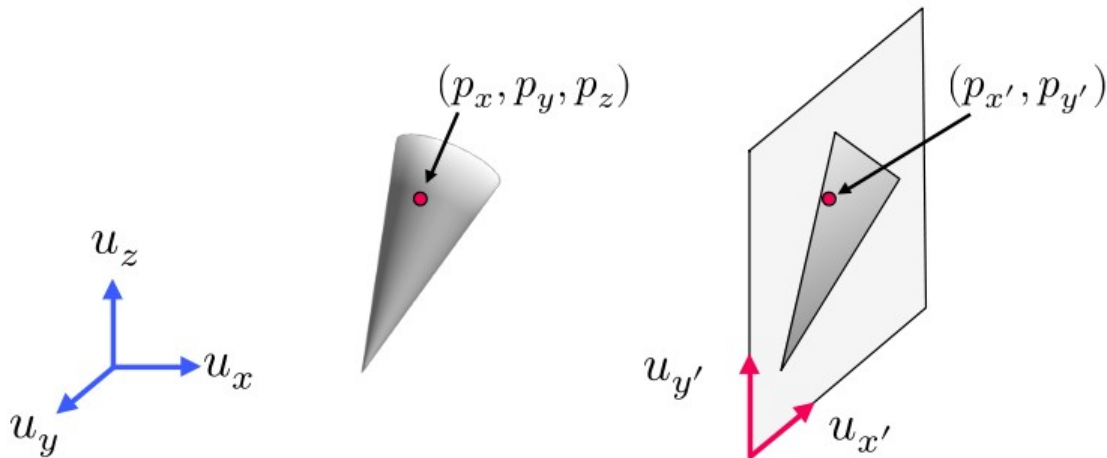
z axis:



$$\begin{bmatrix} p_{x'} \\ p_{y'} \\ p_{z'} \end{bmatrix} = \begin{bmatrix} \cos(\alpha) & \sin(\alpha) & 0 \\ -\sin(\alpha) & \cos(\alpha) & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} p_x \\ p_y \\ p_z \end{bmatrix}$$

3D→2D Projections

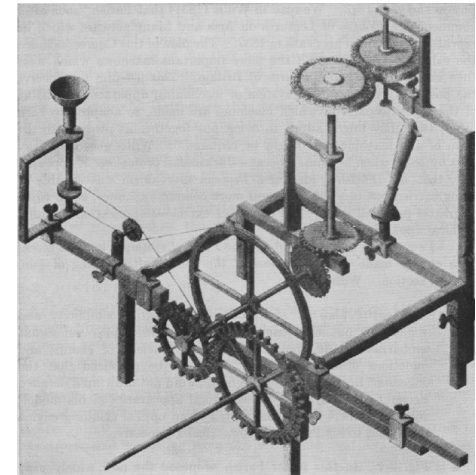
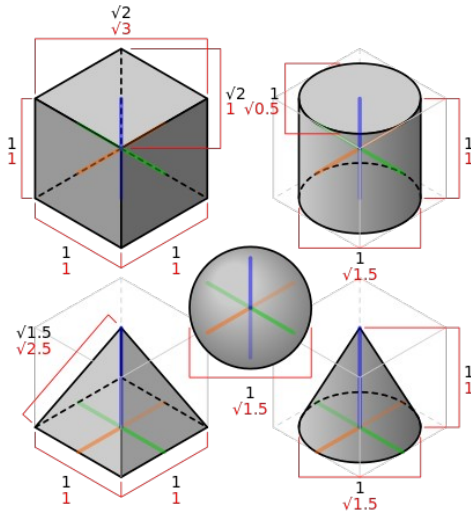
- A **projection** is a coordinate transformation from 3D to 2D.
- Assume that the origin of the two coordinate systems are the same.



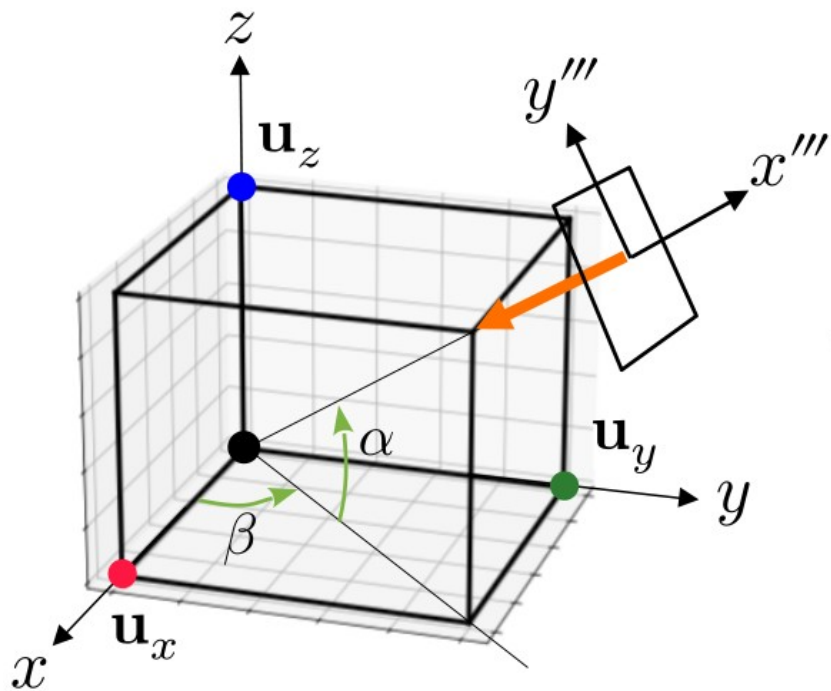
$$\begin{bmatrix} p_{x'} \\ p_{y'} \end{bmatrix} = \begin{bmatrix} 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} p_x \\ p_y \\ p_z \end{bmatrix}$$

Isometric projections

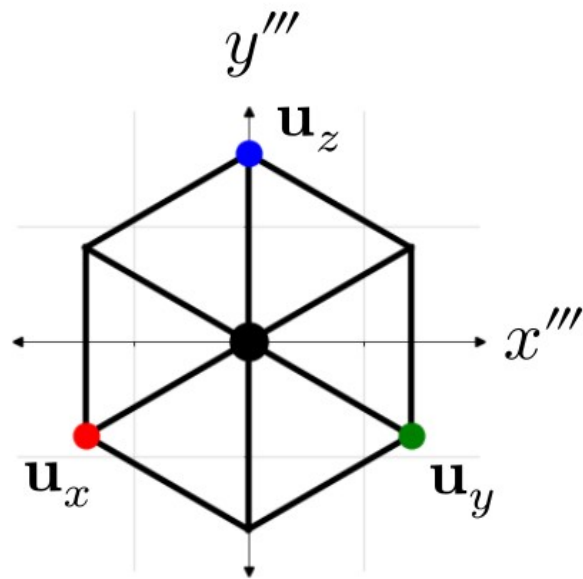
An **isometric projection** is a type of 3D → 2D projection that preserve relative lengths.



World coordinates



Isometric projection



$$\beta = 45^\circ$$

$$\alpha = \text{atan}\left(1/\sqrt{2}\right) \approx 35.264^\circ$$

Isometric projection as a linear transformation

1. Rotation of β about z axis. $(x, y, z) \rightarrow (x', y', z')$

$$\begin{aligned} \begin{bmatrix} p_{x'} \\ p_{y'} \\ p_{z'} \end{bmatrix} &= \begin{bmatrix} \cos(\beta) & \sin(\beta) & 0 \\ -\sin(\beta) & \cos(\beta) & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} p_x \\ p_y \\ p_z \end{bmatrix} \\ &= \underbrace{\begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ -1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{R_1} \begin{bmatrix} p_x \\ p_y \\ p_z \end{bmatrix} \end{aligned}$$

2. Rotation of $-\alpha$ about y' axis. $(x', y', z') \rightarrow (x'', y'', z'')$

$$\begin{bmatrix} p_{x''} \\ p_{y''} \\ p_{z''} \end{bmatrix} = \begin{bmatrix} \cos(\alpha) & 0 & \sin(\alpha) \\ 0 & 1 & 0 \\ -\sin(\alpha) & 0 & \cos(\alpha) \end{bmatrix} \begin{bmatrix} p_{x'} \\ p_{y'} \\ p_{z'} \end{bmatrix} = \underbrace{\begin{bmatrix} \sqrt{2}/\sqrt{3} & 0 & 1/\sqrt{3} \\ 0 & 1 & 0 \\ -1/\sqrt{3} & 0 & \sqrt{2}/\sqrt{3} \end{bmatrix}}_{R_2} \begin{bmatrix} p_{x'} \\ p_{y'} \\ p_{z'} \end{bmatrix}$$

3. Projection: $y'' \rightarrow x'''$, $z'' \rightarrow y'''$

$$\begin{bmatrix} p_{x'''} \\ p_{y'''} \end{bmatrix} = \underbrace{\begin{bmatrix} & \\ & \end{bmatrix}}_{R_3} \begin{bmatrix} p_{x''} \\ p_{y''} \\ p_{z''} \end{bmatrix}$$

Put them together:

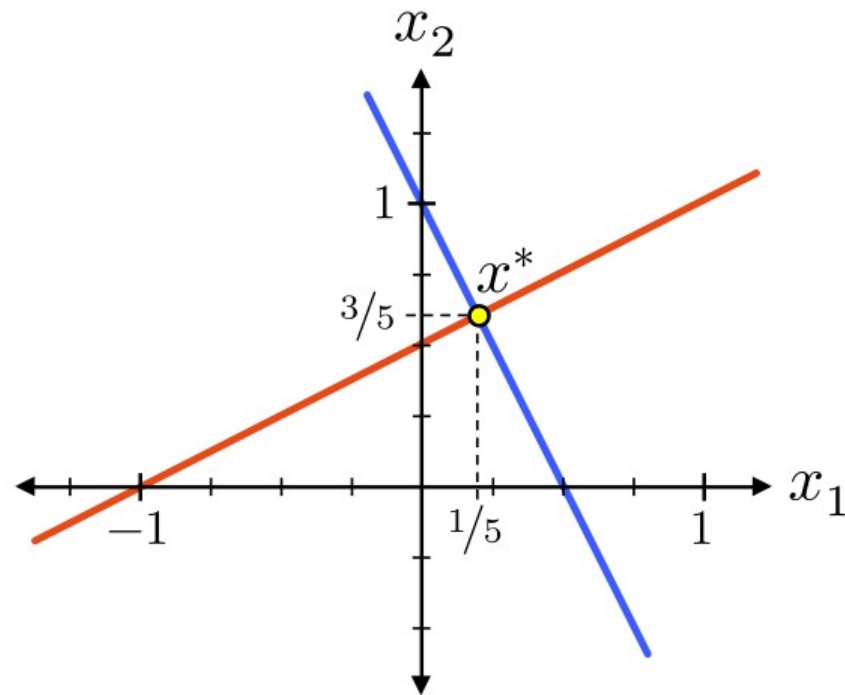
$$\begin{bmatrix} p_{x'''} \\ p_{y'''} \end{bmatrix} = R_3 \begin{bmatrix} p_{x''} \\ p_{y''} \\ p_{z''} \end{bmatrix} = R_3 R_2 \begin{bmatrix} p_{x'} \\ p_{y'} \\ p_{z'} \end{bmatrix} = \underbrace{R_3 R_2 R_1}_M \begin{bmatrix} p_x \\ p_y \\ p_z \end{bmatrix}$$

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Lecture week 13-B **Systems of linear equations**

System of linear equations

$$\begin{cases} 2x_1 + x_2 = 1 & \text{blue line} \\ -x_1 + 2x_2 = 1 & \text{red line} \end{cases}$$



Solution:

$$x^* = (x_1, x_2) = \left(\frac{1}{5}, \frac{3}{5} \right)$$

System of equations in function (a.k.a. matrix) form

The *matrix form* of a system of linear m equations is n unknowns:

$$Ax = b$$

The diagram illustrates the matrix form of a system of linear equations $Ax = b$. It shows the dimensions of the matrix A , the vector x , and the vector b .

- Matrix A is represented by a grid of dots. A vertical double-headed arrow to its left is labeled m , indicating m rows. A horizontal double-headed arrow above it is labeled n , indicating n columns.
- Vector x is shown as a column vector with elements $x_1, \vdots, x_n$. A vertical double-headed arrow to its right is labeled n , indicating n elements.
- Vector b is shown as a column vector with elements $\cdot, \cdot, \cdot, \cdot, \cdot$. A vertical double-headed arrow to its right is labeled m , indicating m elements.

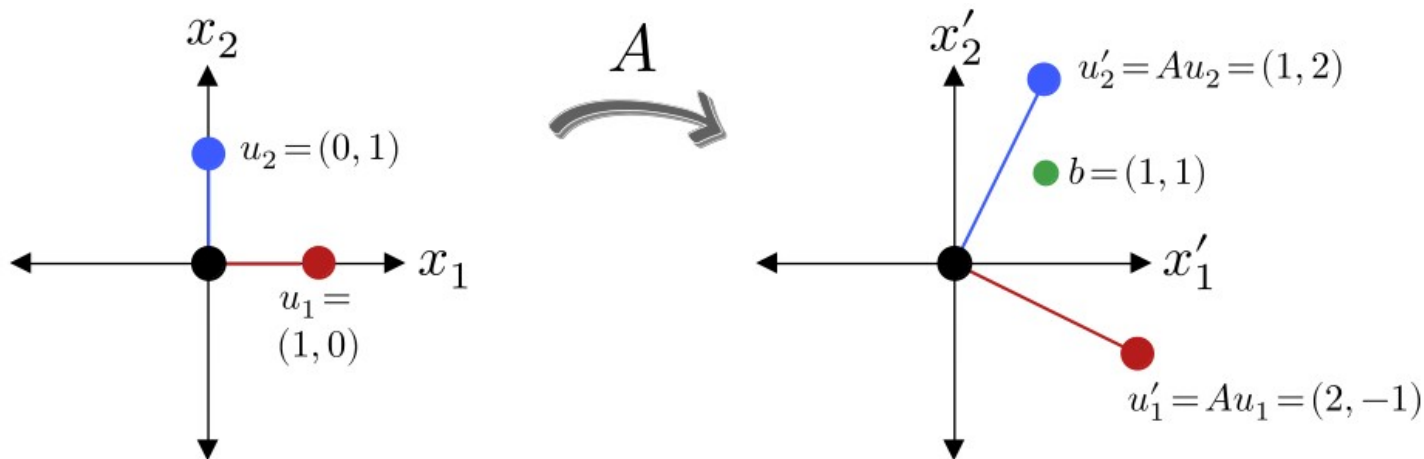
The equation is written as $Ax = b$, with the dimensions of each term clearly indicated by the arrows.

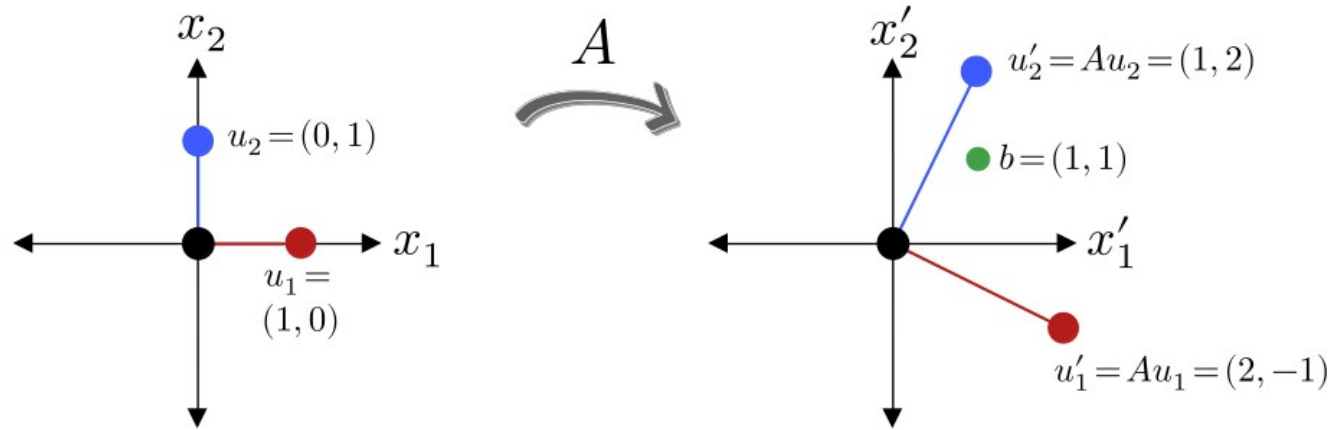
Example Put the following system of equations into matrix form.

$$\begin{cases} 2x_1 + x_2 = 1 \\ -x_1 + 2x_2 = 1 \end{cases}$$

$$\begin{bmatrix} 2 & 1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Function (or geometric) perspective:



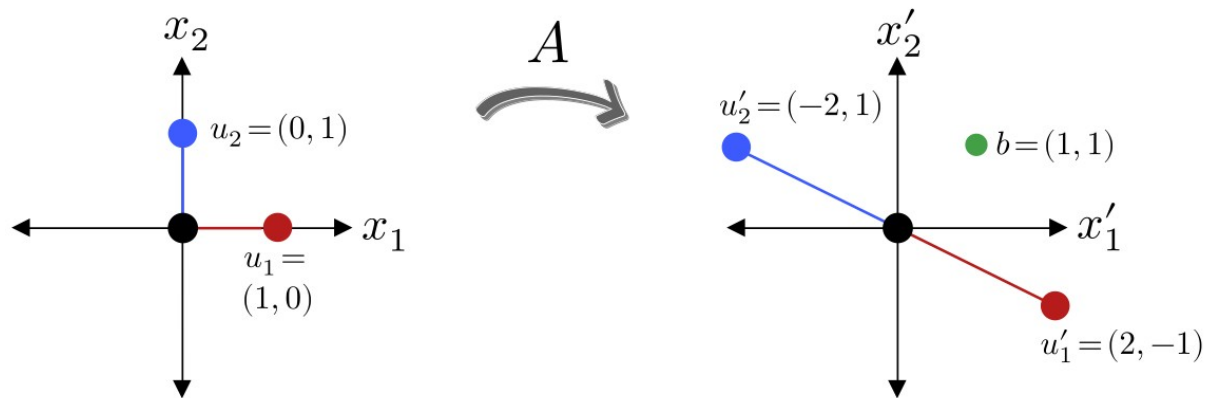


- *Span* of A :
 - The points of the codomain that are reachable via A .
 - The points that can be constructed as a combination of u'_1 and u'_2 .
- *Rank* of A : The dimension of the span of A .

NumPy : `rankA = np.matrix_rank(A)`

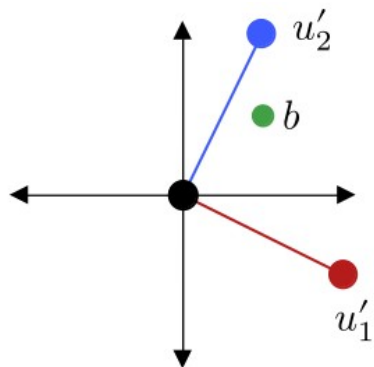
Example Two equations and two unknowns with no solution.

$$\begin{cases} 2x_1 - 2x_2 = 1 \\ -x_1 + x_2 = 1 \end{cases} \qquad \begin{bmatrix} 2 & -2 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$



- u'_1 and u'_2 are colinear. Therefore rank of A equals 1.
- b is not in the span of A . Therefore there are no solutions.

Three cases

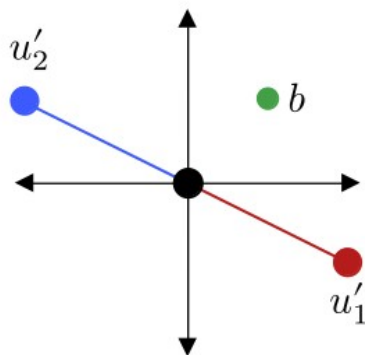


$$\text{rank}(A) = n$$



1 solution

well-determined



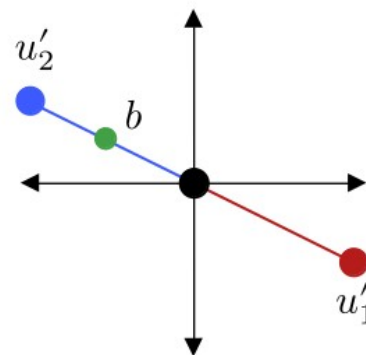
$$\text{rank}(A) < n$$

$$b \notin \text{span}(A)$$



0 solutions

overdetermined



$$\text{rank}(A) < n$$

$$b \in \text{span}(A)$$



∞ solutions

underdetermined

Augmented matrix $A|b$

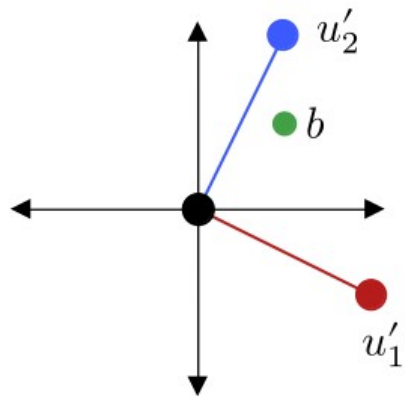
- The *augmented matrix* $A|b$ is obtained by horizontally stacking A and b .

$$A = \begin{bmatrix} 2 & -2 \\ -1 & 1 \end{bmatrix} \quad b = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$A|b = \begin{bmatrix} 2 & -2 & 1 \\ -1 & 1 & 1 \end{bmatrix}$$

NumPy : `Ab = np.hstack((A,b))`

The three cases in terms of the augmented matrix



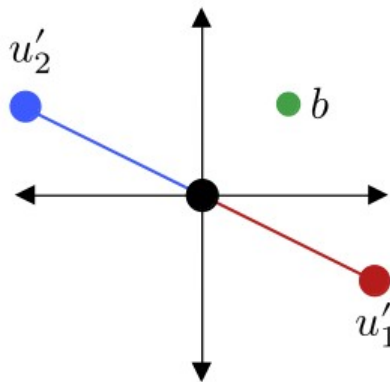
$$\text{rank}(A|b) = \text{rank}(A)$$

$$= n$$



1 solution

well-determined

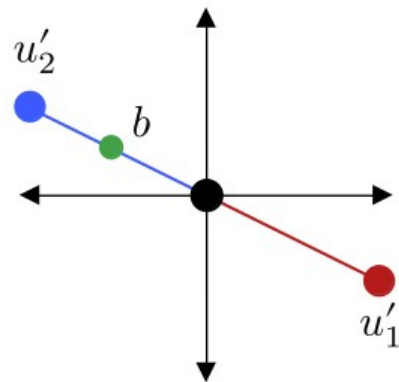


$$\text{rank}(A|b) > \text{rank}(A)$$



0 solutions

overdetermined



$$\text{rank}(A|b) = \text{rank}(A)$$

$$< n$$



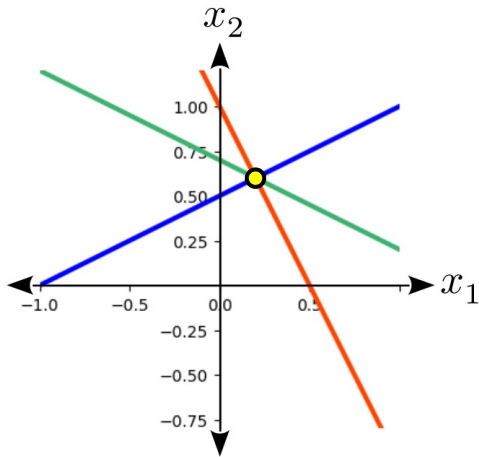
∞ solutions

underdetermined

Examples of equations, two unknowns $m = 3, n = 2$

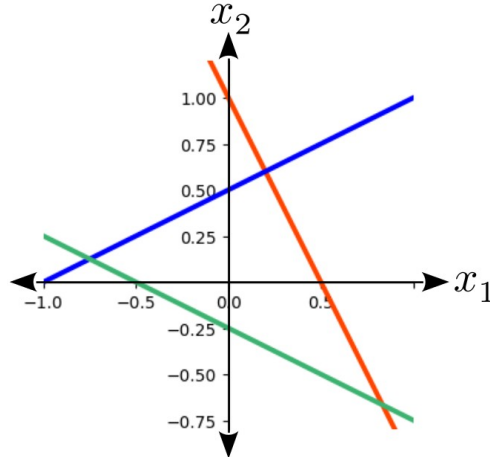
well-determined
(1 solution)

$$\begin{cases} 2x_1 + x_2 = 1 \\ -x_1 + 2x_2 = 1 \\ x_1 + 2x_2 = 1.4 \end{cases}$$



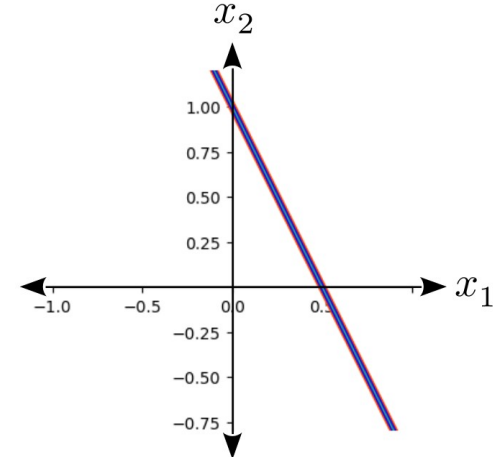
overdetermined
(0 solutions)

$$\begin{cases} 2x_1 + x_2 = 1 \\ -x_1 + 2x_2 = 1 \\ x_1 + 2x_2 = -0.5 \end{cases}$$



underdetermined
(∞ solutions)

$$\begin{cases} 2x_1 + x_2 = 1 \\ -2x_1 - x_2 = -1 \\ 4x_1 + 2x_2 = 2 \end{cases}$$

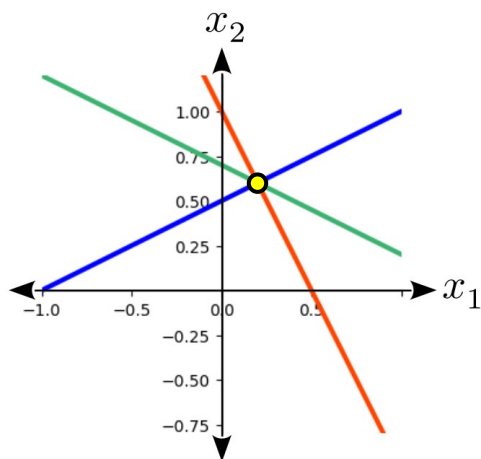


well-determined

$$A|b = \begin{bmatrix} 2 & 1 & 1 \\ -1 & 2 & 1 \\ 1 & 2 & 1.4 \end{bmatrix}$$

$$\text{rank}(A|b) = 2$$

$$\text{rank}(A) = 2$$

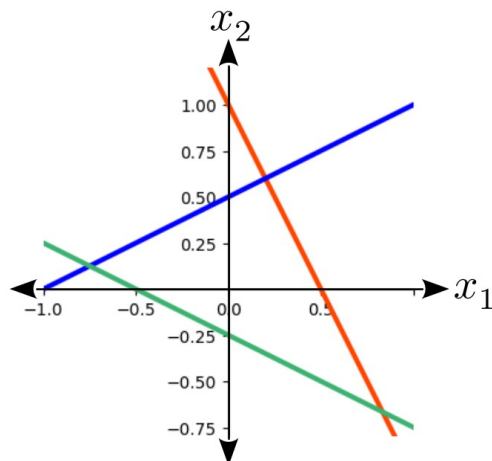


overdetermined

$$A|b = \begin{bmatrix} 2 & 1 & 1 \\ -1 & 2 & 1 \\ 1 & 2 & -0.5 \end{bmatrix}$$

$$\text{rank}(A|b) = 3$$

$$\text{rank}(A) = 2$$

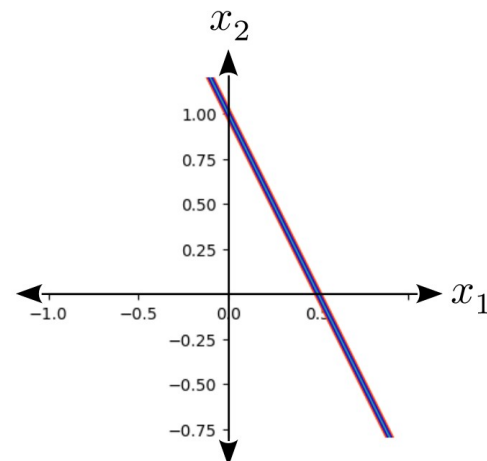


underdetermined

$$A|b = \begin{bmatrix} 2 & 1 & 1 \\ -2 & -1 & -1 \\ 4 & 2 & 2 \end{bmatrix}$$

$$\text{rank}(A|b) = 1$$

$$\text{rank}(A) = 1$$



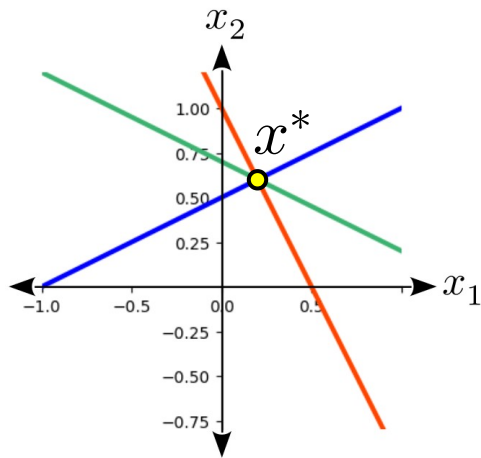
Moore-Penrose pseudoinverse A^+

$$x^* = A^+ b$$

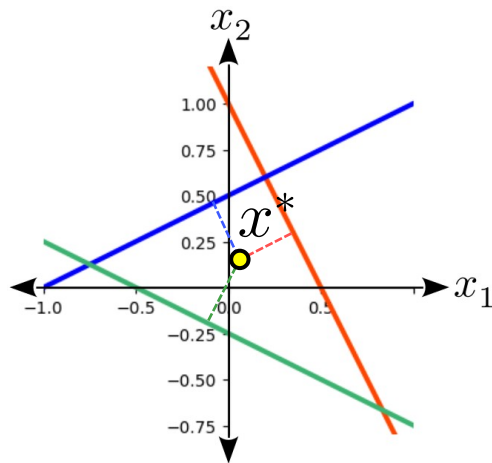
NumPy :

```
Ainv = np.pinv(A)
```

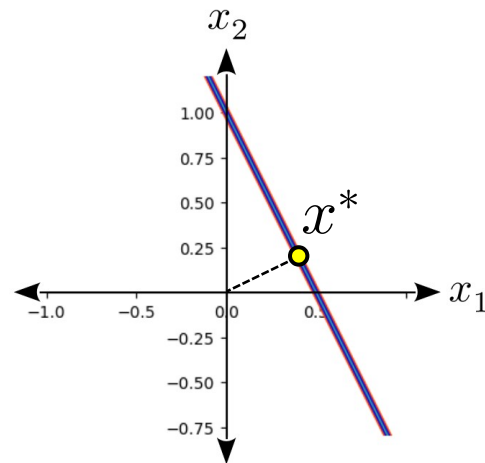
well-determined
(unique solution)



overdetermined
(least squares solution)



underdetermined
(minimum norm solution)

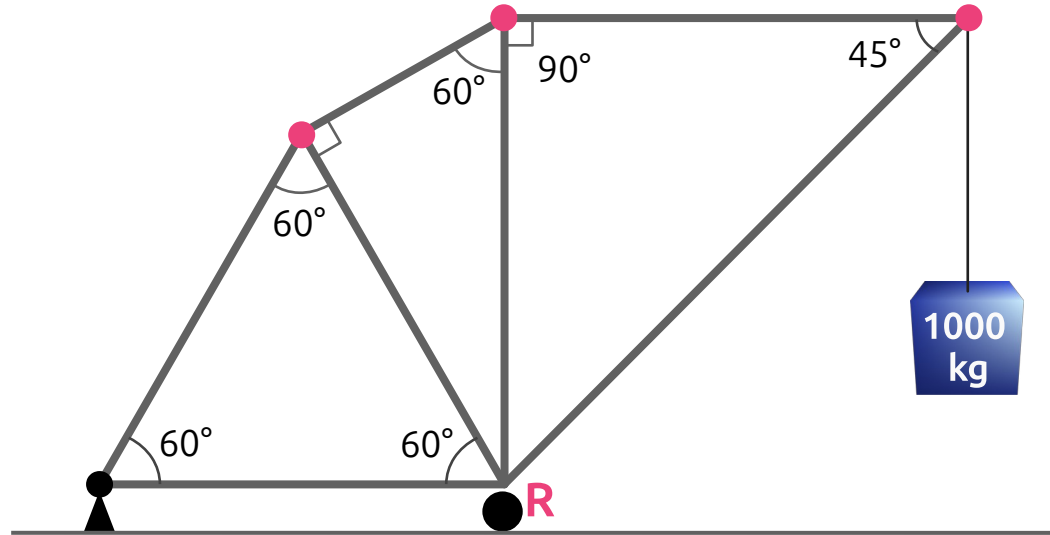


E7 Introduction to Computer Programming for Scientists and Engineers

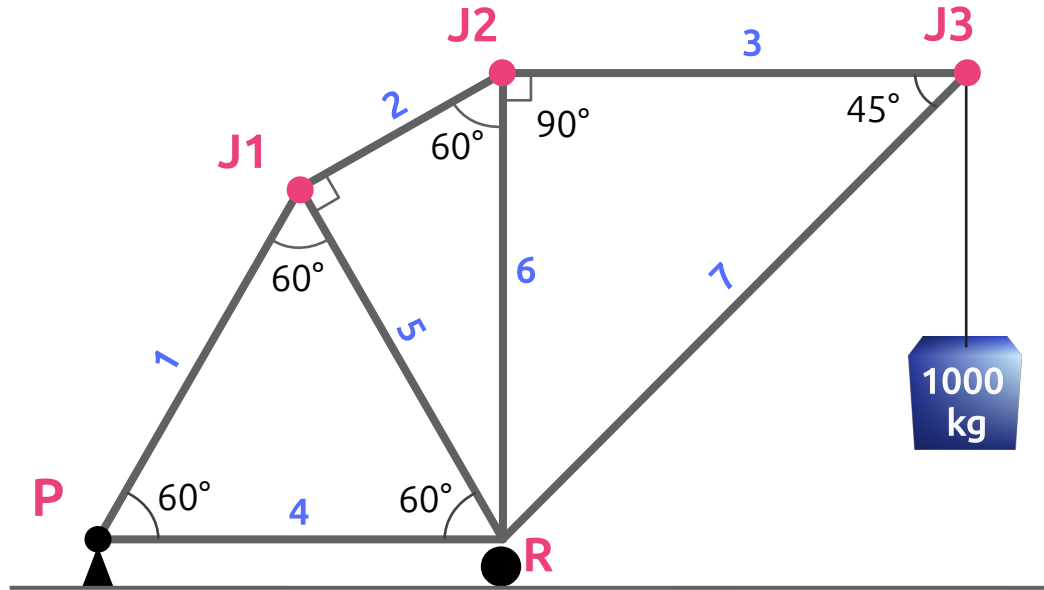
Discussion week 13

**Using linear algebra to
solve the forces in a crane**

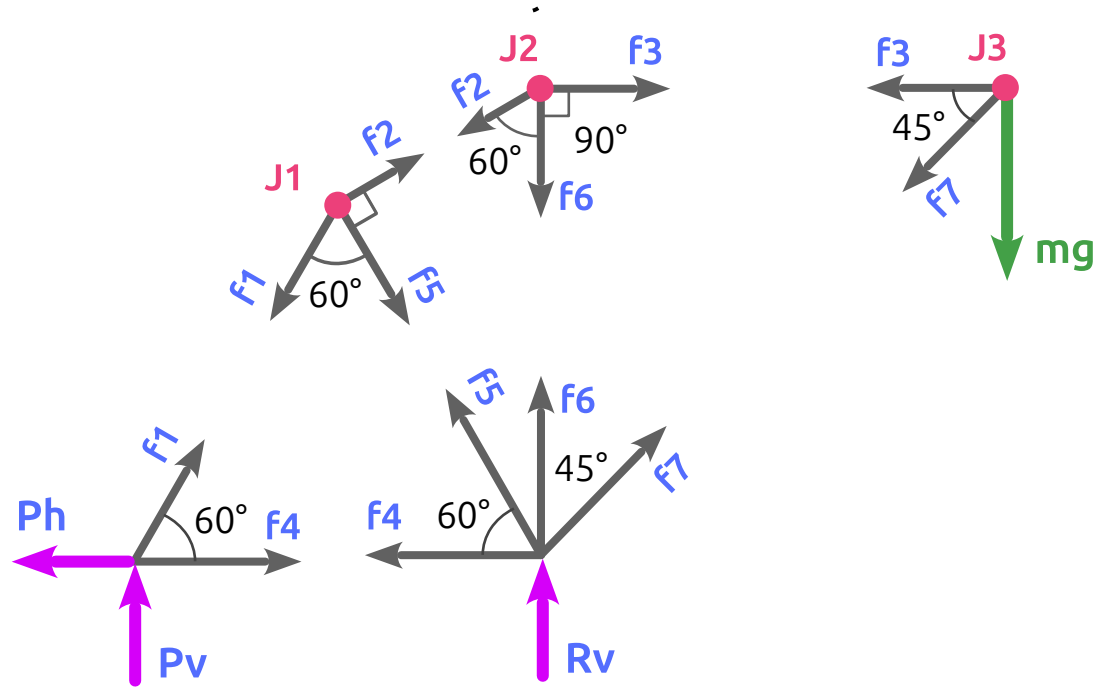
Given the crane shown below, compute the forces present in each of its segments when it lifts a box weighing 1000kg.



Step 1: Label the links and joints.

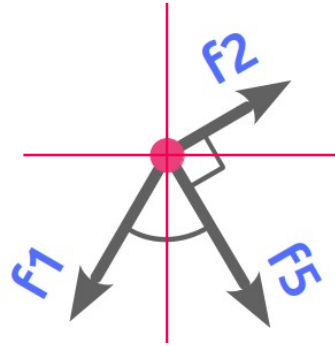


Step 2: Free body diagram for each joint



Step 3: Vertical and horizontal force balance equations

Example **J1**:



System of linear equations

$$J1(h): \sin(30)*f1 - \cos(30)*f2 - \sin(30)*f5 = 0$$

$$J1(v): \cos(30)*f1 - \sin(30)*f2 + \cos(30)*f5 = 0$$

$$J2(h): \cos(30)*f2 - f3 = 0$$

$$J2(v): \sin(30)*f2 + f6 = 0$$

$$J3(h): f3 + \cos(45)*f7 = 0$$

$$J3(v): -\sin(45)*f7 = mg$$

$$P(h): -\sin(30)*f1 - f4 + Ph = 0$$

$$P(v): -\cos(30)*f1 + Pv = 0$$

$$R(h): f4 + \sin(30)*f5 - \cos(45)*f7 = 0$$

$$R(v): -\cos(30)*f5 - f6 - \sin(45)*f7 + Rv = 0$$

Step 4: Put the system of equations into matrix form.

0

J1 (h):

1

J1 (v):

2

J2 (h):

3

J2 (v):

4

J3 (h):

5

J3 (v):

6

P (h):

7

P (v):

8

R (h):

9

R (v):

0

f1

1

f2

2

f3

3

f4

4

f5

5

f6

6

f7

7

Ph

8

Pv

9

Rv

s30	-c30			-s30					
c30	-s30			c30					
	c30	-1							
	s30				1				
		1				c45			
						-s45			
-s30			-1				1		
-c30								1	
			1	s30		-c45			
				-c30	-1	-s45			1

f1

f2

f3

f4

f5

f6

f7

Ph

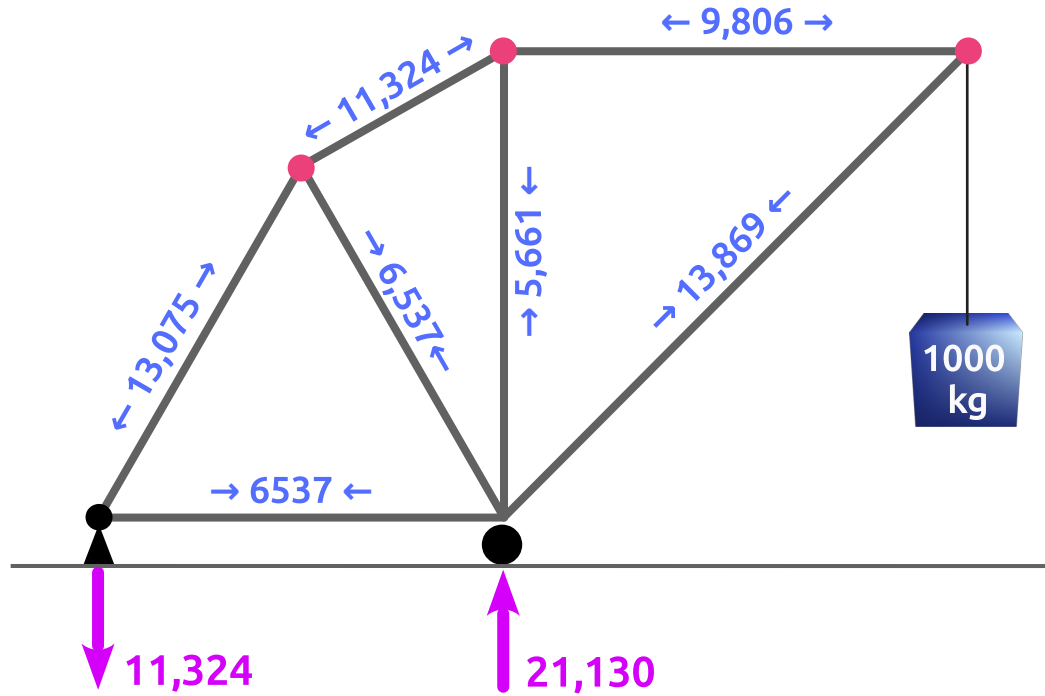
Pv

Rv

=

mg

Step 5: Solve in code.



E7 Introduction to Computer Programming for Scientists and Engineers

Lecture week 14-A/B

Numerical methods for solving initial values problems

Recap We've covered numerical methods for solving *algebraic* equations.

- Single non-linear equations such as $x^4 - 3x^3 + x^2 + 1 = 0$

- _____

- _____

- _____

- Systems of non-linear equations such as $\begin{cases} x^2 - y = 1 \\ y^2 - x = 1 \end{cases}$

- _____

- Systems of linear equations such as the "forces in a crane" problem.

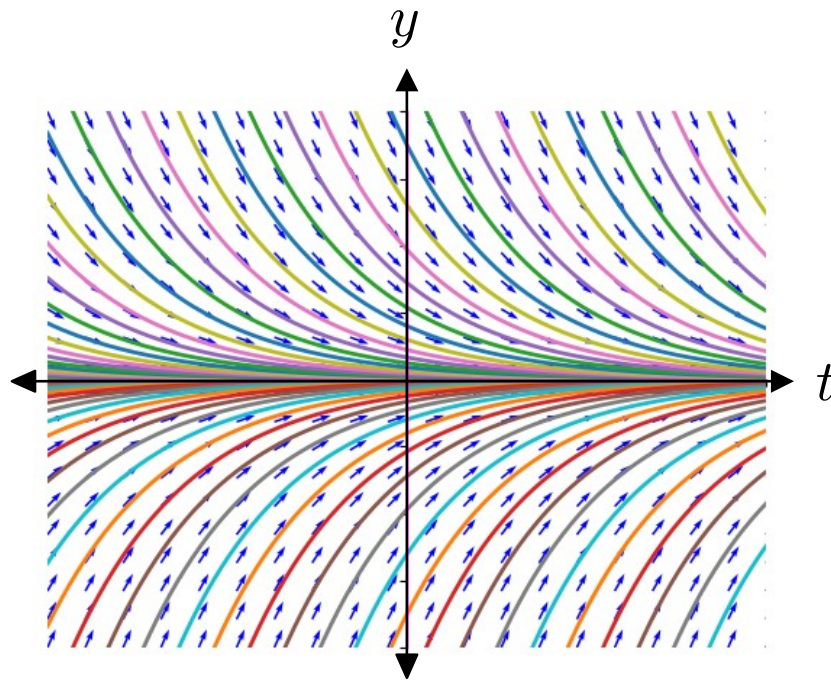
- _____

Ordinary differential equation (ODE)

An ODE is an equation whose unknown is a ***function*** $y : \mathbb{R} \rightarrow \mathbb{R}$

Example Find $y: \mathbb{R} \rightarrow \mathbb{R}$ such that $y' = -y$

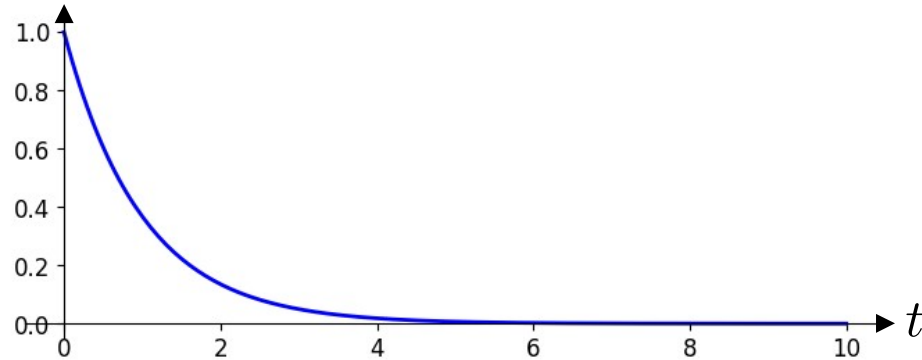
- Solution: $y(t) = a e^{-t}$ $a \in \mathbb{R}$



Initial value problem (IVP)

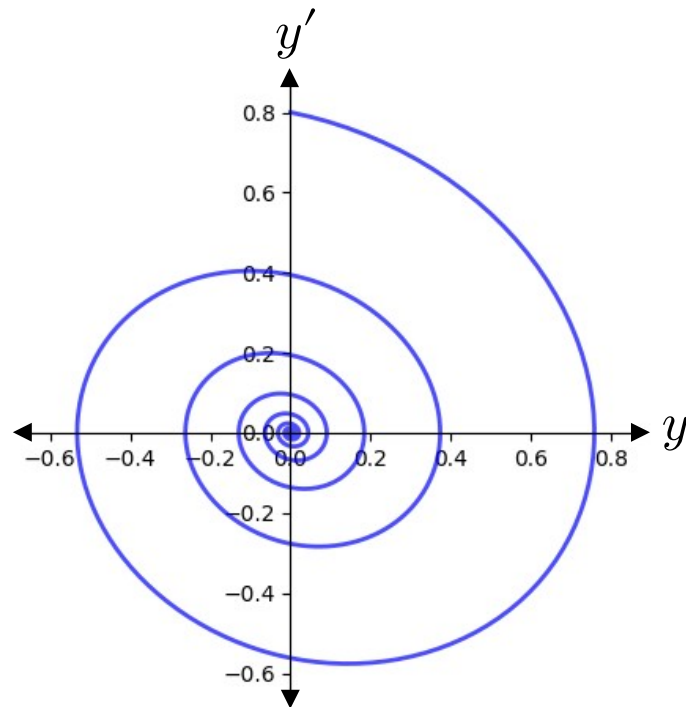
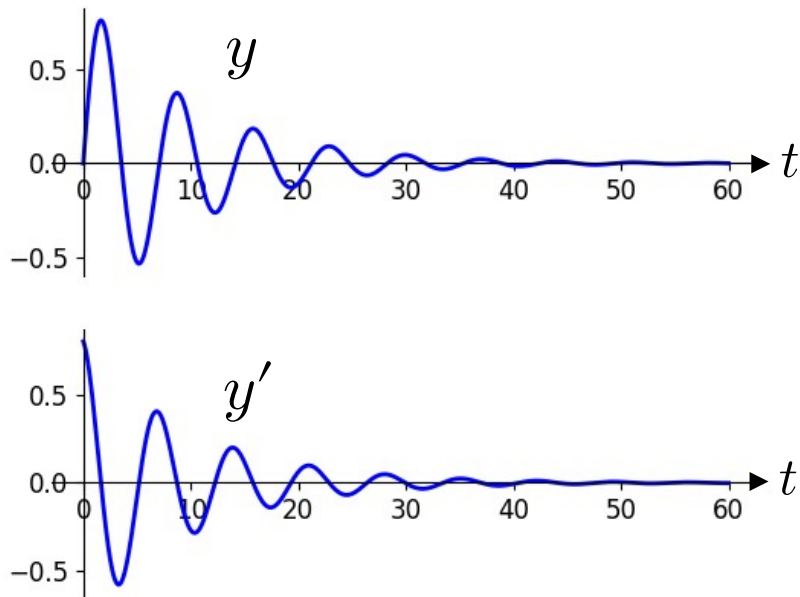
An IVP is an ODE with the value of y at time zero given.

Example $y' = -y$ with $y(0) = 1$



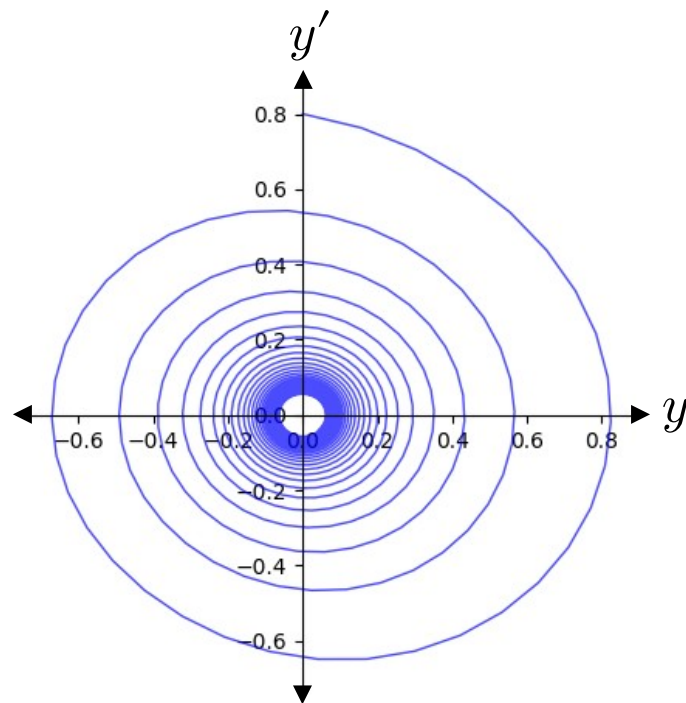
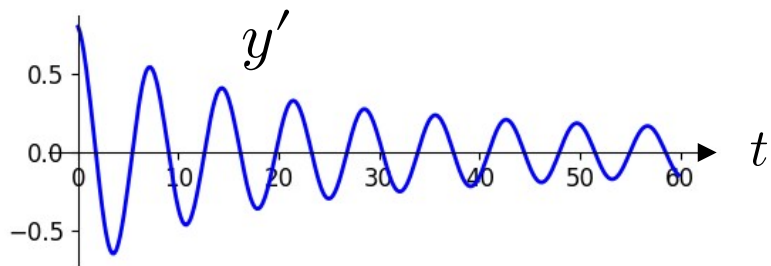
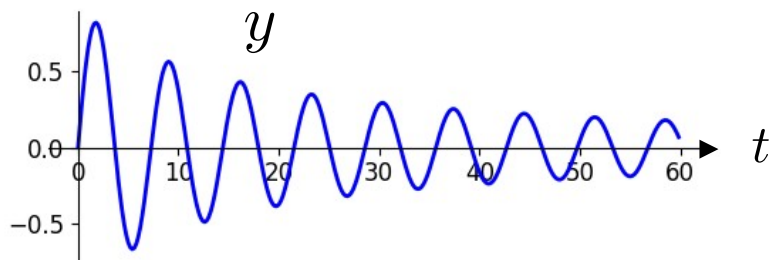
Example Given $a > 0$, $b > 0$, find $y: \mathbb{R} \rightarrow \mathbb{R}$ such that,

$$y'' + a y' + b y = 0$$



Example Given $a > 0$, $b > 0$, find $y: \mathbb{R} \rightarrow \mathbb{R}$ such that,

$$y'' + a y' |y'| + b \sin(y) = 0$$

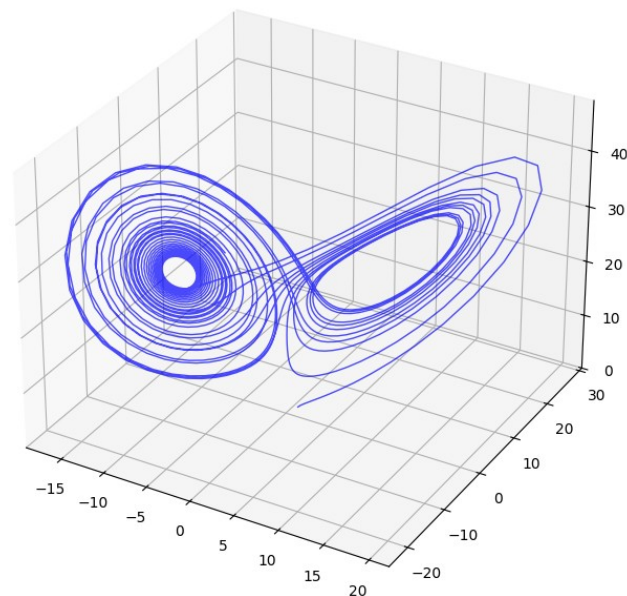
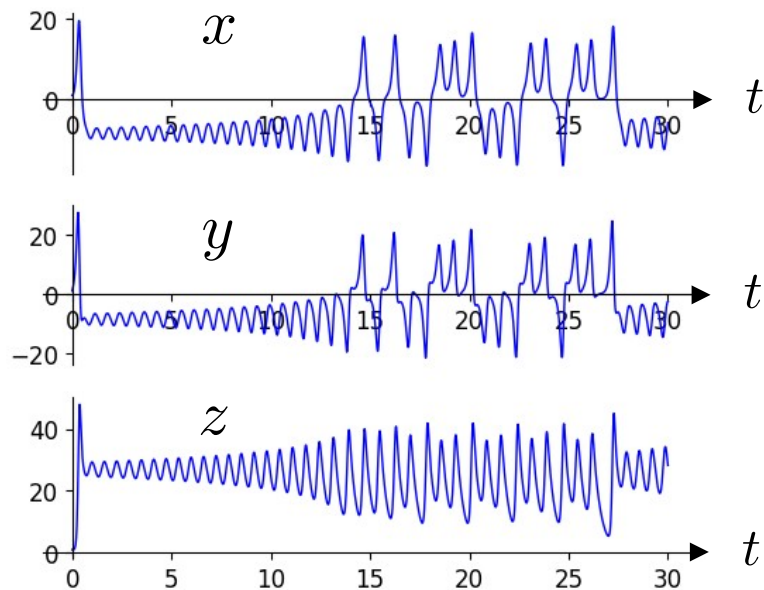


Example Given $\sigma > 0$, $\rho > 0$, $\beta > 0$, find scalar functions $x(t)$, $y(t)$, $z(t)$ such that

$$x' = \sigma(y - x)$$

$$y' = x(\rho - z) - y$$

$$z' = xy - \beta z$$



Solving an IVP numerically

1) If the ODE is of order >1 , put it into **vector form**: $\mathbf{y}' = f(\mathbf{y}, t)$

2) Apply a **numerical method** to the vector form.

- Euler's method
- Midpoint method
- Heun's method
- etc.

Putting an ODE into vector form $\mathbf{y}' = f(\mathbf{y}, t)$

Example: $y'' + a y' + b y = 0$

1) Define n new variables representing y and its first $n - 1$ derivatives.

2) $y'_i = y_{i+1}$ for each i from 1 to $n-1$.

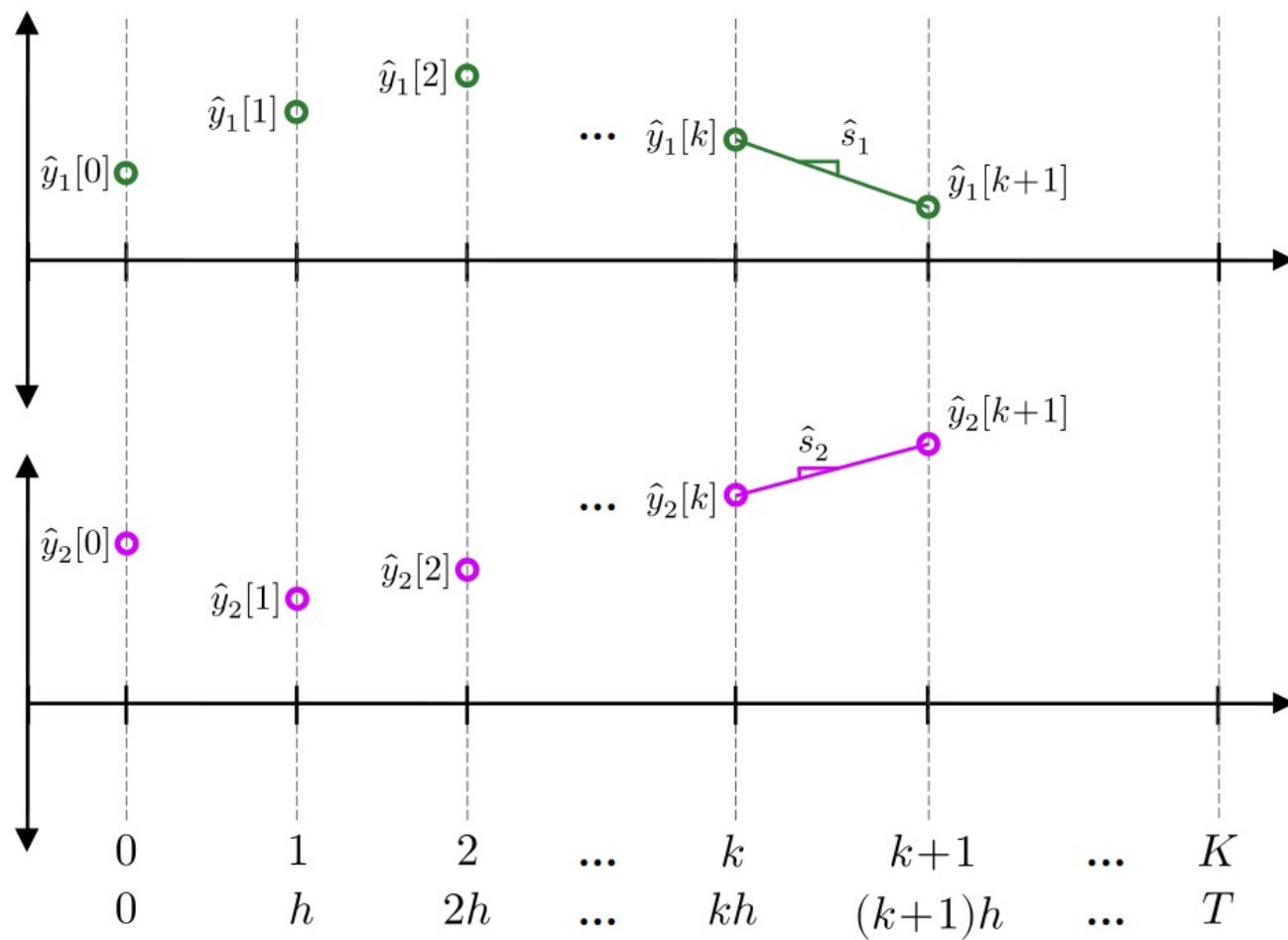
3) Use the ODE to express y'_n in terms of y_1 through y_n .

4) Write the vector form $\mathbf{y}' = f(\mathbf{y}, t)$ with $\mathbf{y} = (y_1, \dots, y_n)$.

Numerical solution of $\mathbf{y}' = f(\mathbf{y}, t)$

- Generic iteration: $\hat{\mathbf{y}}[0] = \mathbf{y}(0)$
$$\hat{\mathbf{y}}[k + 1] = \hat{\mathbf{y}}[k] + \hat{\mathbf{s}} h$$
- $\hat{\mathbf{y}}[k] = (\hat{y}_1[k], \dots, \hat{y}_n[k])$... numerical solution at time step k .
- $\hat{\mathbf{s}} = (\hat{s}_1, \dots, \hat{s}_n)$... vector of slopes.
- Each numerical method computes $\hat{\mathbf{s}}$ differently, based on:
 - f ... the vector state equation
 - h ... the time step
 - $\hat{\mathbf{y}}[k]$... the current estimate
 - t ... the current time

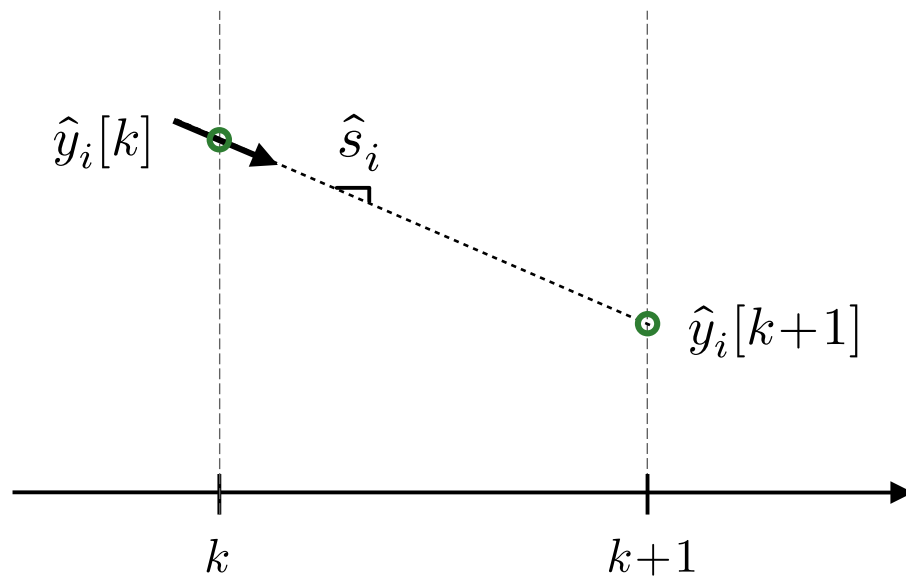
Generic iteration



Euler's method

$$\hat{\mathbf{s}} = f(\hat{\mathbf{y}}[k], t)$$

Plot of i 'th component:



Midpoint method

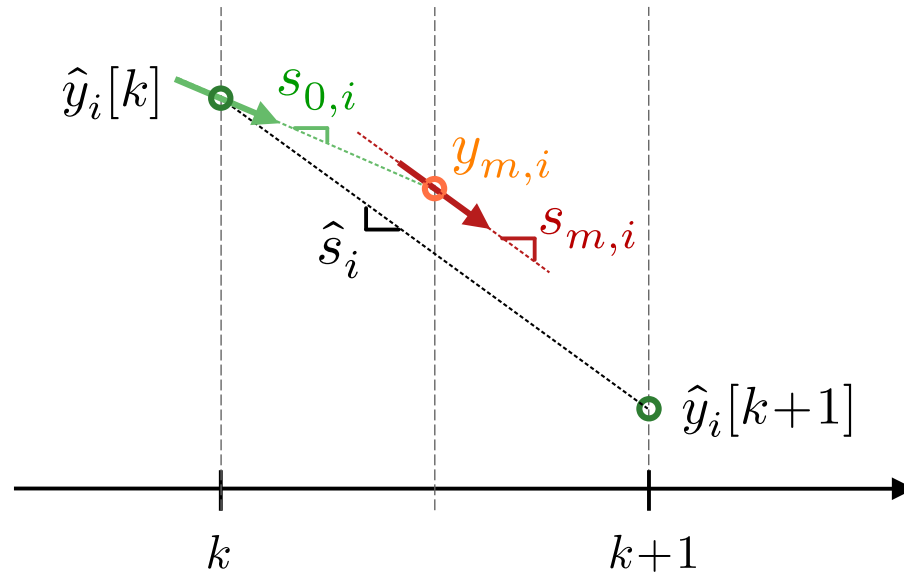
$$\mathbf{s}_0 = f(\hat{\mathbf{y}}[k], t)$$

$$\mathbf{y}_m = \hat{\mathbf{y}}[k] + (h/2)\mathbf{s}_0$$

$$\mathbf{s}_m = f(\mathbf{y}_m, t + h/2)$$

$$\hat{\mathbf{s}} = \mathbf{s}_m$$

Plot of i 'th component:



Heun's method

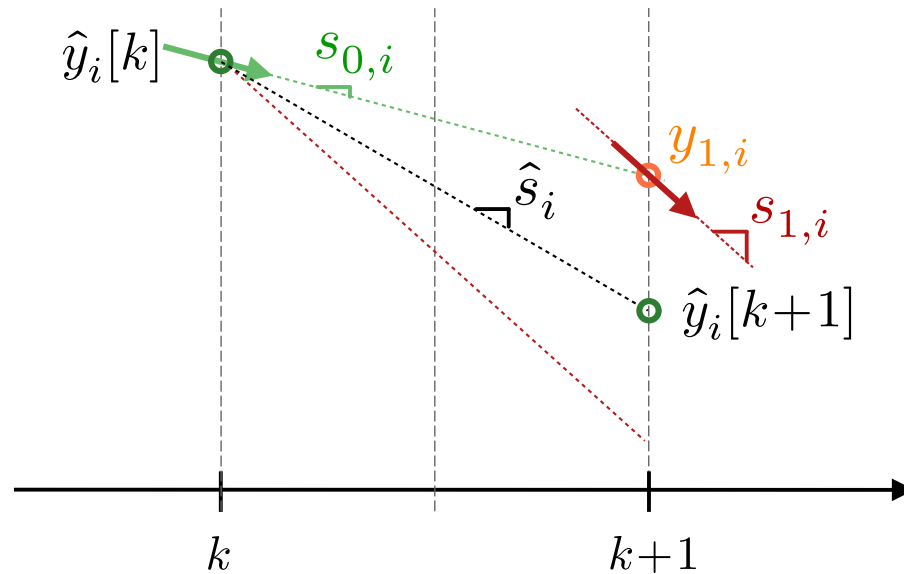
$$\mathbf{s}_0 = f(\hat{\mathbf{y}}[k], t)$$

$$\mathbf{y}_1 = \hat{\mathbf{y}}[k] + h \mathbf{s}_0$$

$$\mathbf{s}_1 = f(\mathbf{y}_1, t+h)$$

$$\hat{\mathbf{s}} = 0.5(\mathbf{s}_0 + \mathbf{s}_1)$$

Plot of i 'th component:



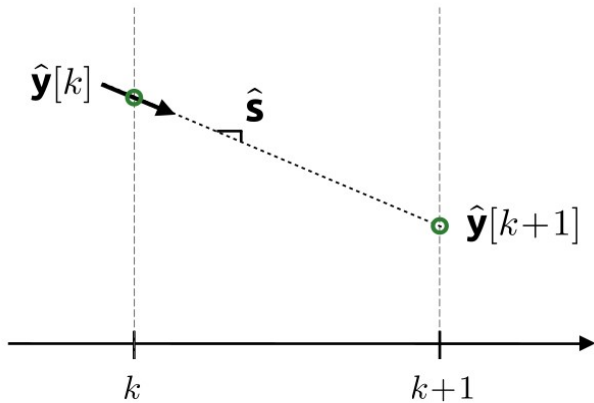
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Lecture week 14-B

Numerical methods for ODEs **(continued)**

Euler's method

$$\hat{\mathbf{s}} = f(\hat{\mathbf{y}}[k], t)$$



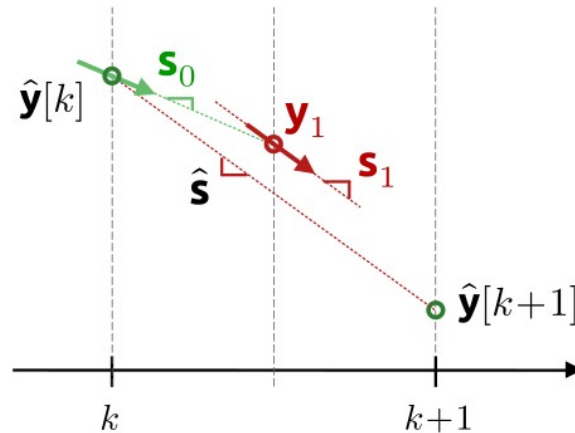
Midpoint method

$$\mathbf{s}_0 = f(\hat{\mathbf{y}}[k], t)$$

$$\mathbf{y}_1 = \hat{\mathbf{y}}[k] + (h/2)\mathbf{s}_0$$

$$\mathbf{s}_1 = f(\mathbf{y}_1, t + h/2)$$

$$\hat{\mathbf{s}} = \mathbf{s}_1$$



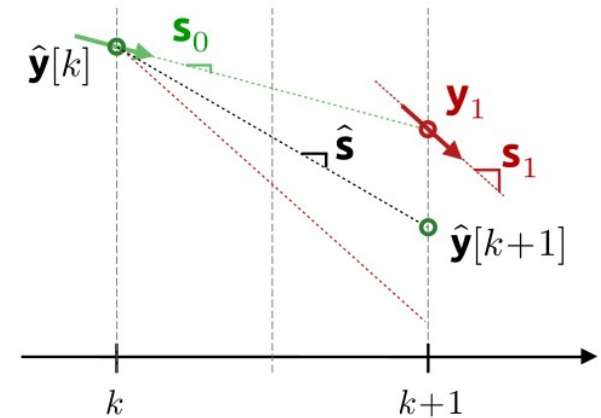
Heun's method

$$\mathbf{s}_0 = f(\hat{\mathbf{y}}[k], t)$$

$$\mathbf{y}_1 = \hat{\mathbf{y}}[k] + h\mathbf{s}_0$$

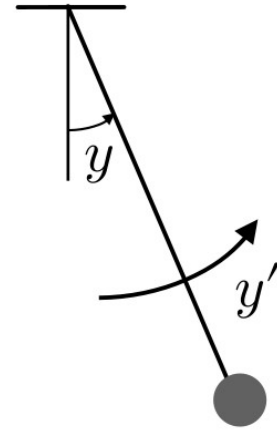
$$\mathbf{s}_1 = f(\mathbf{y}_1, t + h)$$

$$\hat{\mathbf{s}} = 0.5(\mathbf{s}_0 + \mathbf{s}_1)$$



Example: Simple pendulum

$$y'' + ay'|y'| + b \sin(y) = 0$$



Runge-Kutta 4

$$\mathbf{s}_0 = f(\hat{\mathbf{y}}[k], t)$$

$$\mathbf{y}_1 = \hat{\mathbf{y}}[k] + (h/2) \mathbf{s}_0$$

$$\mathbf{s}_1 = f(\mathbf{y}_1, t + h/2)$$

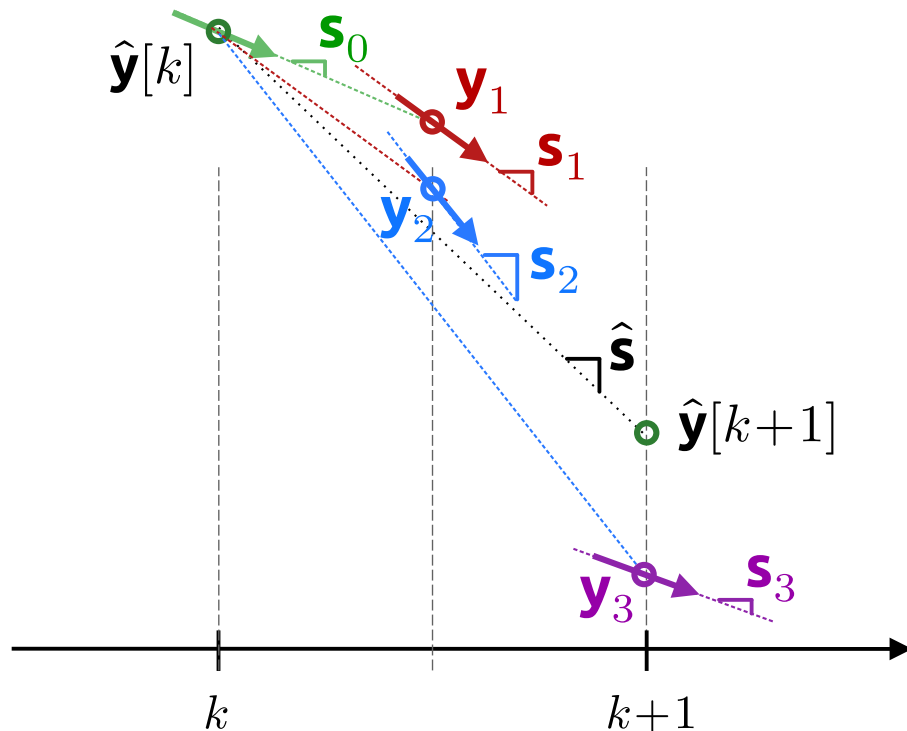
$$\mathbf{y}_2 = \hat{\mathbf{y}}[k] + (h/2) \mathbf{s}_1$$

$$\mathbf{s}_2 = f(\mathbf{y}_2, t + h/2)$$

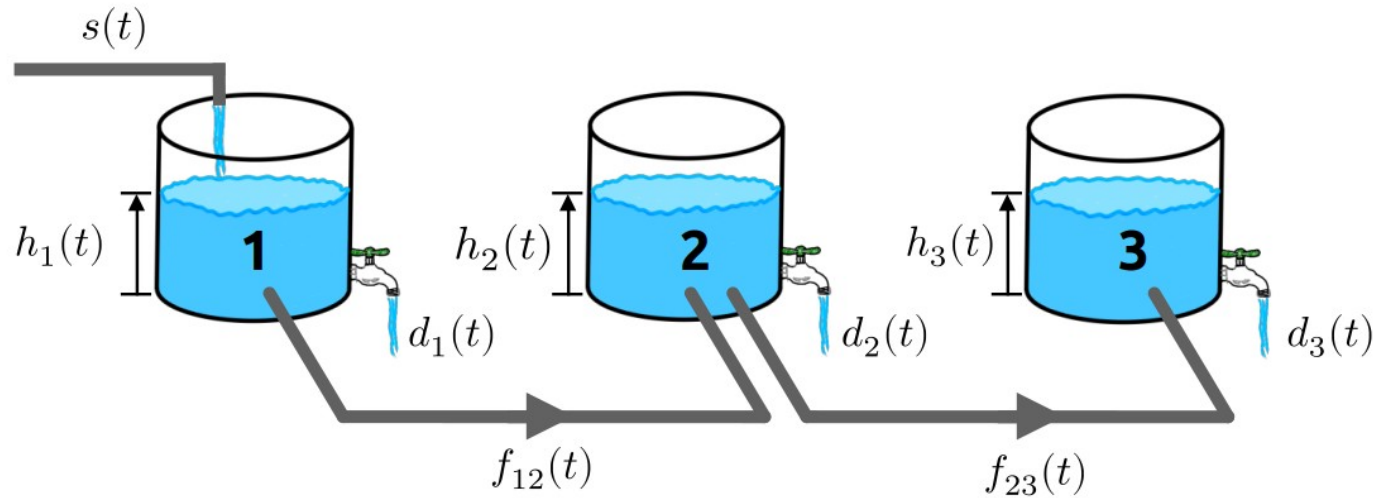
$$\mathbf{y}_3 = \hat{\mathbf{y}}[k] + h \mathbf{s}_2$$

$$\mathbf{s}_3 = f(\mathbf{y}_3, t + h)$$

$$\hat{\mathbf{s}} = (1/6) (\mathbf{s}_0 + 2 \mathbf{s}_1 + 2 \mathbf{s}_2 + \mathbf{s}_3)$$



Example: Interconnected water tanks



An automatic water balancing system sets the flows between the tanks to:

$$f_{12}(t) = \alpha_1(h_1(t) - h_2(t))$$

$$f_{23}(t) = \alpha_2(h_2(t) - h_3(t))$$

E7 Introduction to Computer Programming for Scientists and Engineers

Discussion week 14

Randomness and Monte Carlo simulation



Berkeley
ENGINEERING

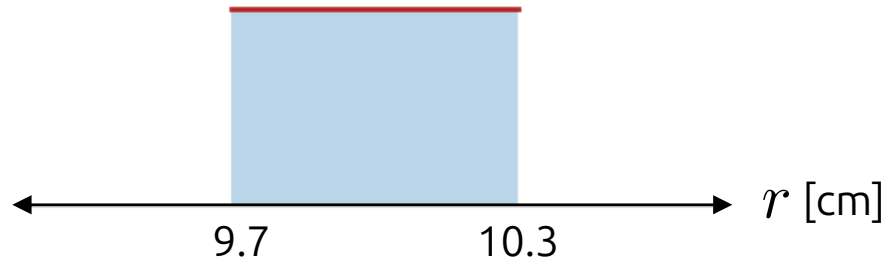
Systems we've encountered

System	Inputs	Solve	To obtain
spherical mixer	h	evaluate a function	total volume
heated plate	measured temperatures	system of linear equations	unmeasured temperatures
simple pendulum	$\ell, m, y(0), y'(0)$	IVP	energy after 20 swings
three tanks	$a, h_i(0), s(t), d_i(t)$	IVP	$\max_{t \in [0,1]} f_{12}(t)$

Modeling uncertainty with probability theory

- Random variable: A variable whose value is ***sampled*** from a ***distribution***.

Example: The radius r of the spherical mixer is *uniformly distributed* between 9.7 cm and 10.3 cm.

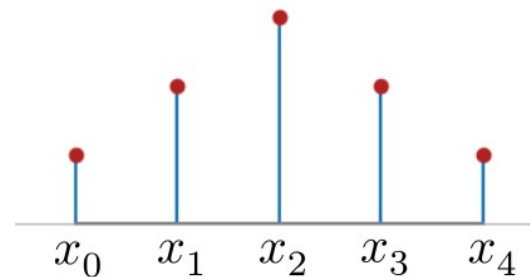


- A random variable has a **sample space** and a **distribution**.
- The **sample space** of a random variable is the set of all possible values.
- The **distribution** of a random variable is a function $p: X \rightarrow \mathbb{R}$ that encodes the sampling probabilities.
- The distribution has these properties:
 - Positivity: $p(x) \geq 0 \quad \forall x \in X$
 - Unit integral (or sum):

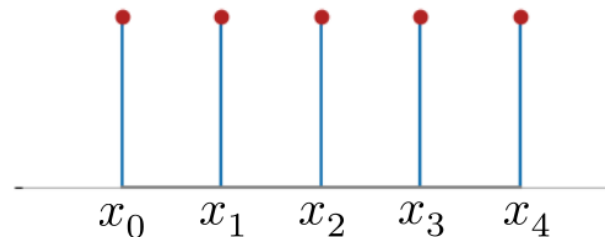
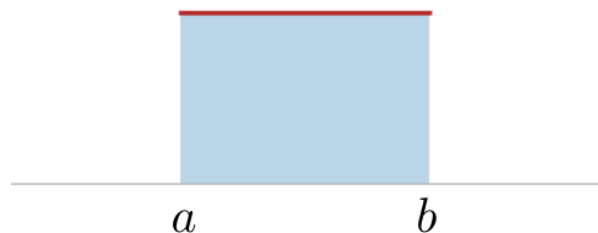
$$\int_X p(x) dx = 1 \quad \text{or} \quad \sum_{x \in X} p(x) = 1 \quad \text{if } X \text{ is discrete}$$

Some important types of random variables

- **Discrete r.v. :** X is a finite set.

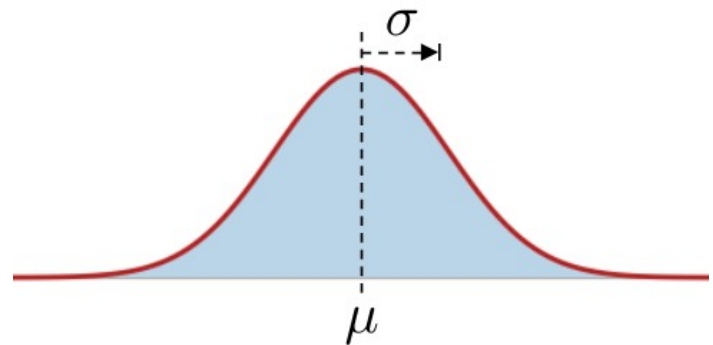


- **Uniform r.v. :** $p(x)$ is constant over X .



- **Gaussian (a.k.a. normal) r.v. :**

- $X = \mathbb{R}$.
- $p(x)$ is “bell-shaped”.

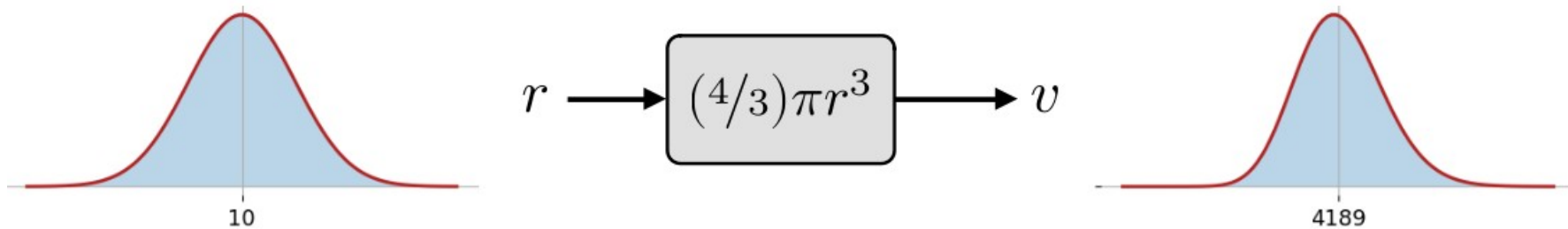


Functions of random variables

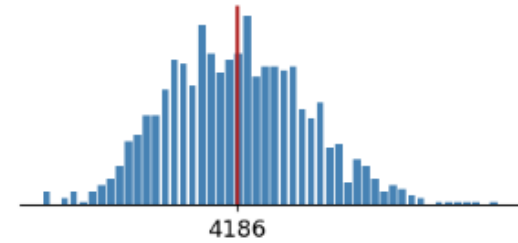
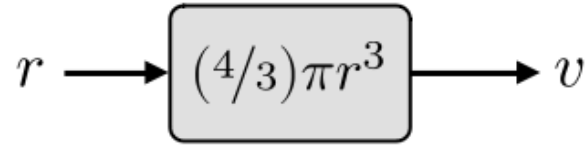
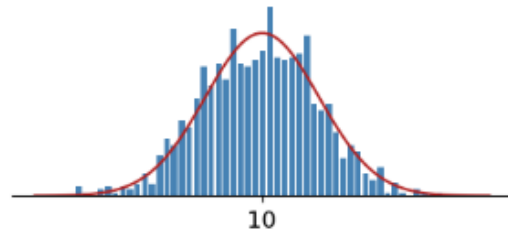
- The radius r of the spherical mixer is modeled as a uniform distribution between 97 mm and 103 mm.
- The total volume v of the sphere relates to r through

$$v = \left(\frac{4}{3}\right)\pi r^3$$

- What is the distribution of v ?



Monte Carlo simulation



In general

samples of
input values



samples of
output values

Sampling distributions with SciPy



- Create a random variable with a given distribution:

- Generic discrete `X = scipy.stats.rv_discrete(values=(X,p))`
- Continuous uniform `X = scipy.stats.uniform(loc=a,scale=b-a)`
- Gaussian `X = scipy.stats.norm(loc=mu,scale=sigma)`

- Sample the random variable:

```
X.rvs(size=<int>)
```

```
X.rvs(size=<int>,random_state=<int>)
```

... to ensure repeatability.

E7 Introduction to Computer Programming for Scientists and Engineers

Lecture week 15-A **Randomness**

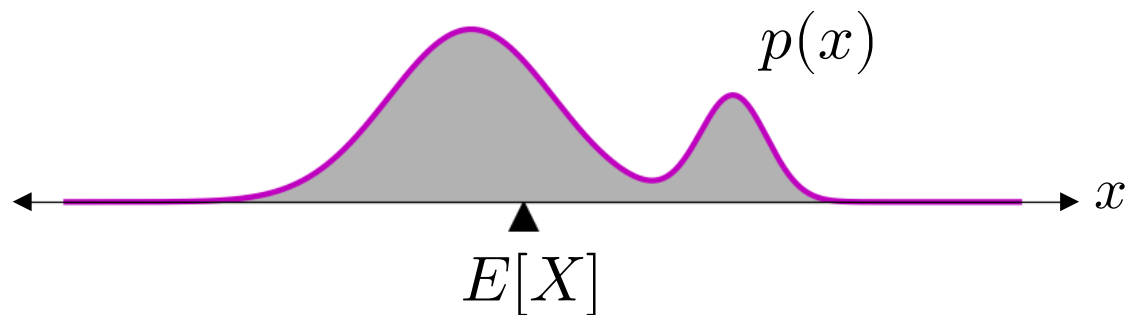
Mean of a random variable (a.k.a. *expected value*)

- X is a random variable with sample space Ω and distribution $p(x)$.
- The **mean** of X , denoted $E[X]$ or μ_X , is a real number defined with

$$E[X] = \int_{\Omega} x p(x) dx \quad \dots \text{if } \Omega \text{ is continuous.}$$

$$E[X] = \sum_{x \in \Omega} x p(x) \quad \dots \text{if } \Omega \text{ is discrete.}$$

Example: Mean of a continuous r.v.



Example: Mean of a discrete r.v.

x	$p(x)$
-1	0.2
0	0.3
2	0.5

Standard deviation of a random variable

- X is a random variable with sample space Ω and distribution $p(x)$.
- The **standard deviation** of X , denoted $\text{Std}[X]$ or σ_X , is a real number defined with

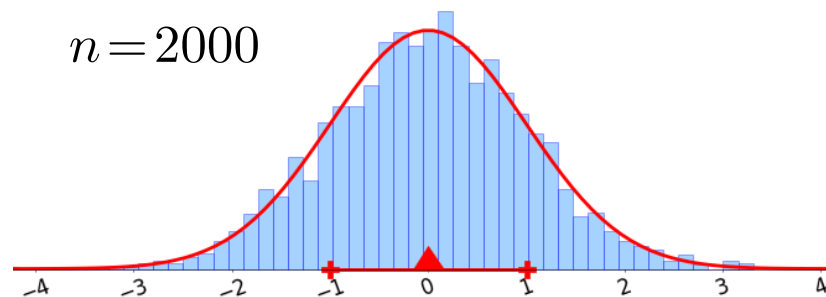
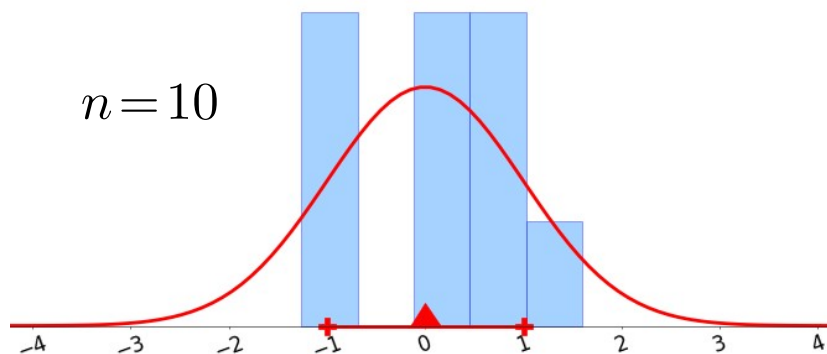
$$\text{Std}[X] = \sqrt{E[(X - \mu_X)^2]}$$

Estimating the mean and standard deviation from samples

- We have a dataset $\mathcal{D}$ consisting of n samples of X .

$$\mathcal{D} = \{x_1, \dots, x_n\}$$

- The histogram of $\mathcal{D}$ approaches the distribution of X as $n \rightarrow \infty$.



- Estimate the mean with the **sample mean** $\hat{\mu}_n$.

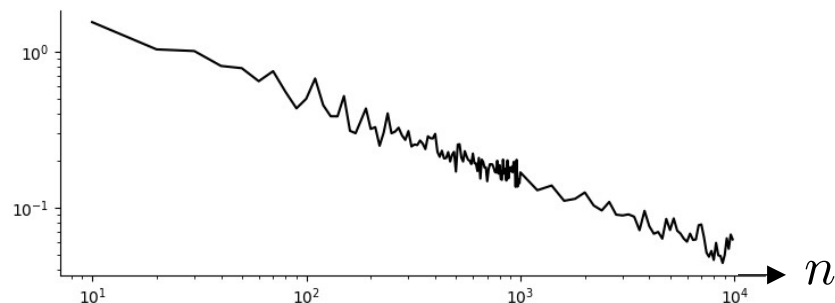
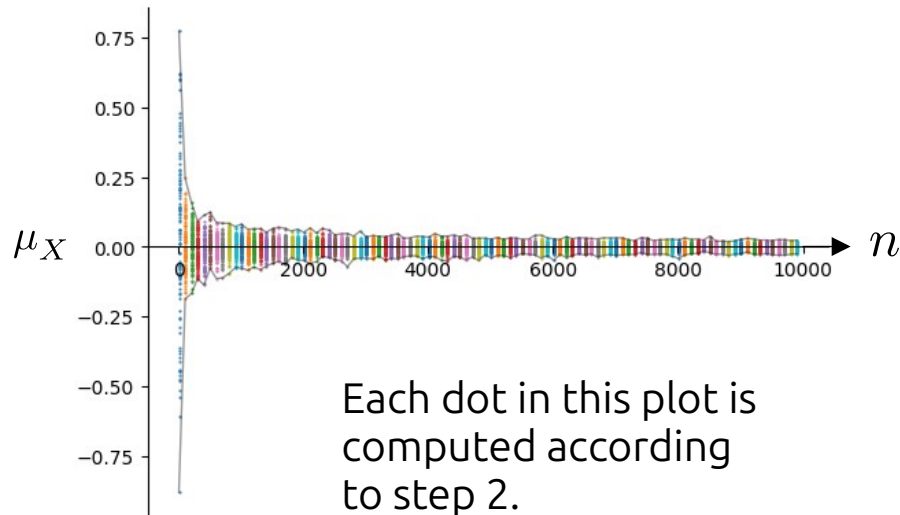
$$\hat{\mu}_n = \frac{1}{n} \sum_{i=1}^n x_i$$

- Estimate the standard deviation with the **sample standard deviation** $\hat{\sigma}_n$.

$$\hat{\sigma}_n = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \hat{\mu}_n)^2}$$

Convergence of the sample mean to the true mean

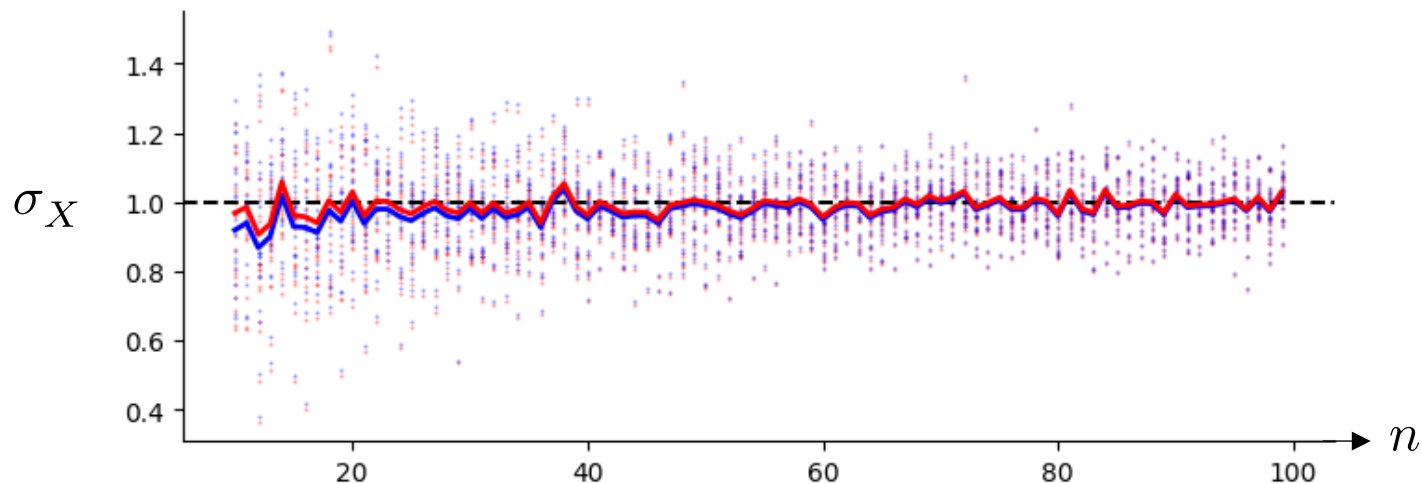
1. We have a random variable X with $\mu_X = 0$.
2. We collect a sample $\mathcal{D}$ with n samples and calculate $\hat{\mu}_n$.
3. Repeat this 100 times for each n .
4. Do this for increasing values of n .



This plot shows the variation in the 100 samples of $\hat{\mu}_n$ computed in step 3.

Justification for the “n-1” in the definition of $\hat{\sigma}_n$

- Red dots are samples of $\hat{\sigma}_n$ defined with “ $n - 1$ ”.
- Blue dots are samples of $\hat{\sigma}_n$ defined with “ n ”.
- The red and blue lines are the averages of the dots.
- The blue line underestimates the true value on average.



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Lecture week 15-B

Randomness - continued

Calculating the probabilities of events from distributions

- X is a random variable with sample space Ω and distribution p .
- An **event** of X is any subset of its sample space: $e \subseteq \Omega$
- The probability that event e occurs is the integral (or sum) of $p(x)$ over e .

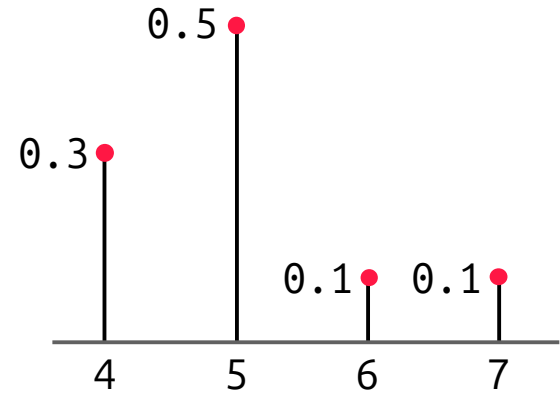
Continuous r.v.

$$\text{Prob}(e) = \int_e p(x) dx$$

Discrete r.v.

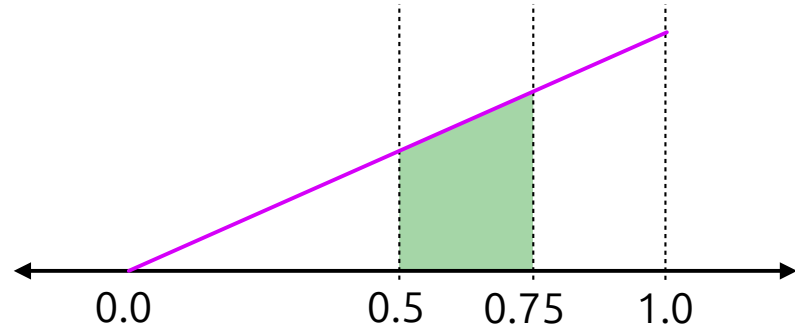
$$\text{Prob}(e) = \sum_{x \in e} p(x)$$

Example For a discrete r.v. with distribution shown below, what is the probability of the event $\{4,6\}$?



Example

For the continuous random variable with distribution show below, what is the probability of the event $[0.5, 0.75]$?



Estimating the probabilities of events from samples

Given a dataset $\mathcal{D}$ consisting of n samples of X

$$\mathcal{D} = \{x_1, \dots, x_n\}$$

we can estimate the probability of an event e with

$$\hat{P}(e) = \frac{\# \text{ samples in } e}{n}$$

Example

Given the dataset plotted below, estimate the probability of the event $[0.5, 0.75]$

